ICPC Notebook

common

build.sh

```
g++ a.cpp -o a -O2 -std=c++20 -fsanitize=address 2>err.txt &&
./a <in.txt >out.txt 2>err.txt
```

clangd.md

.clangd

インクルードパスの確認は `gcc -v -c -xc++`

```
InlayHints:
  Enabled: No
CompileFlags:
  Add:
    - -std=gnu++20
    - -Wall
    - -I/usr/include/c++/13
    - -I/usr/include/x86_64-linux-gnu/c++/13
```

hash_line.sh

```
# 標準入力から C++ ファイルを受け取り、コメント・空白・改行を削除し、md5
でハッシュする
y=""
g++ -dD -E -P -fpreprocessed - | while IFS= read x; do
y=$y$x
echo $(echo "$y" | tr -d '[:space:]' | md5sum | cut -c-2) "$x"
done
```

template.hpp

md5: e98612

```
2b  #include <bits/stdc++.h>
e0  #include <cassert>
32  using namespace std;
3b  using ll = long long;
84  const ll INF = 1ll << 60;
64  #define REP(i, n) for (ll i=0; i<ll(n); i++)
35  template <class T> using V = vector<T>;
23  template<class A, class B>
c9  bool chmax(A &a, B b) { return a < b && (a = b, true); }
f4  template<class A, class B>
e9  bool chmin(A &a, B b) { return b < a && (a = b, true); }
```

structure

persistent_avl.hpp

md5: 3d3a63

```
3c  template <int BUFSIZE, class S, S op(S, S), S e(), class F,
de            S mapping(F, S), F comp(F, F), F id()>
06  struct Node {
78    using np = Node*;
d9    using pnp = pair<np, np>;
87    np l, r;
```

```
be    S sum;
3a    F lz;
ca    int dep;
1e    static inline Node buf[BUFSIZE];
36    static inline int pbuf = 0;
c6    static np alloc() { return buf + pbuf++; }
0c    static int mdep(np a, np b) {
a6      return max(a->dep, b->dep) + 1;
45    }
65    static np mid(np l, np r, F lz = id()) {
a0      auto v = alloc();
97      *v = {l, r, mapping(lz, op(l->sum, r->sum)), lz,
1e            mdep(l, r)};
d3      return v;
2b    }
c0    np apply(F f) {
a2      if (l) return mid(l, r, comp(f, lz));
6a      return &(*alloc() = {0, 0, mapping(f, sum), id(), 0});
55    }
82    static pnp mg1(np a, np b, np c) {
58      if (abs(mdep(a, b) - c->dep) <= 1)
67        return {mid(a, b), c};
90      if (abs(mdep(b, c) - a->dep) <= 1)
b6        return {a, mid(b, c)};
f5      return {mid(a, b->l->apply(b->lz)),
35              mid(b->r->apply(b->lz), c)};
e1    }
c2    static pnp mg2(np a, np b, F alz, F blz) {
21      if (a->dep < b->dep) {
26        blz = comp(blz, b->lz);
7c        auto [xl, xr] = mg2(a, b->l, alz, blz);
dd        return mg1(xl, xr, b->r->apply(blz));
8a      }
5d      if (a->dep > b->dep) {
61        alz = comp(alz, a->lz);
7a        auto [xl, xr] = mg2(a->r, b, alz, blz);
4c        return mg1(a->l->apply(alz), xl, xr);
f1      }
3d      return {a->apply(alz), b->apply(blz)};
f6    }
7e    static np merge(np a, np b) {
ac      if (!a || !b) return a ? a : b;
f3      auto [l, r] = mg2(a, b, id(), id());
e9      return mid(l, r);
3e    }
34    pnp split(const auto& det, S off = e(), F f = id()) {
bf      if (det(op(off, sum))) return {apply(f), 0};
fc      if (!l) return {0, apply(f)};
3d      f = comp(f, lz);
c9      if (S t = op(off, l->sum); det(t)) {
ac        auto [gl, gr] = r->split(det, t, f);
49        return {merge(l->apply(f), gl), gr};
a2      }
9f      auto [gl, gr] = l->split(det, off, f);
99      return {gl, merge(gr, r->apply(f))};
47    }
3b    static np build(const vector<S>& arr) {
75      vector<np> tmp;
4b      for (auto v : arr) {
3d        tmp.push_back(alloc());
b3        *tmp.back() = {0, 0, v, id(), 0};
66      }
e4      int n = tmp.size();
af      for (int s = 1; s < n; s *= 2)
ee        for (int i = 0; i + s < n; i += 2 * s)
2a          tmp[i] = merge(tmp[i], tmp[i + s]);
90      return tmp[0];
87    }
3d  };
```

2D_segtree.hpp

md5: 856cb8

```
b6  template <class Coord>
39  struct TwoDRectangleQuery {
b5    int n, lg;
9c    V<V<int>> L, idx;
e3    V<int> Z, rx;
78    V<Coord> xs, ys;
d9    TwoDRectangleQuery() {}
bb    TwoDRectangleQuery(V<pair<Coord, Coord>> pos)
20        : n(pos.size()),
a6          lg(countr_zero(bit_ceil(unsigned(n)))),
d3          L(lg, V<int>(n + 1)), idx(lg + 1, V<int>(n, -1)),
38          Z(lg), rx(n), xs(n), ys(n) {
e7      V<pair<Coord, int>> iy(n), ix(n);
40      REP(i, n) ix[i] = {pos[i].first, i};
79      REP(i, n) iy[i] = {pos[i].second, i};
81      sort(iy.begin(), iy.end());
e6      sort(ix.begin(), ix.end());
b5      REP(i, n) xs[i] = ix[i].first;
6d      REP(i, n) rx[ix[i].second] = i;
1b      REP(i, n) tie(ys[i], idx[lg][i]) = iy[i];
bb      for (int i = lg - 1; i >= 0; i--) {
e4        int h = 1 << i;
83        auto& prei = idx[i + 1];
1f        Z[i] = (n >> (i + 1) << i) + min(h, n % (h * 2));
70        int ai = 0, bi = Z[i];
cc        REP(k, n) {
c6          bool cb = rx[prei[k]] & h;
0f          idx[i][cb ? bi++ : ai++] = prei[k];
b5          L[i][k + 1] = L[i][k] + !cb;
9f        }
a4      }
62      REP(i, n) rx[idx[lg][i]] = i;
6b    }
97    struct Pt{ int d, i; };
16    V<Pt> getUpdatePoints(int v) {
c7      V<Pt> q(lg + 1, {lg, rx[v]});
7a      for (auto [d, p] = q[0]; d--;) {
7c        int h = L[d][p];
4a        q[d] = {d, p = L[d][p + 1] == h ? Z[d] + p - h : h};
31      }
1e      return q;
09    }
5a    struct Range{ int d, l, r; };
2f    void rec(V<Range>& dest, int l, int r, int yl, int yr,
ea             int a, int b, int d) {
fe      if (r <= a || b <= l || yl == yr) return;
85      if (l <= a && b <= r) {
54        dest.push_back({d, yl, yr});
15        return;
e4      }
66      int xm = a + (1 << --d);
0f      int s = L[d][yl], t = L[d][yr];
09      rec(dest, l, r, s, t, a, xm, d);
e2      rec(dest, l, r, Z[d] + yl - s, Z[d] + yr - t, xm, b,
e0          d);
ab    }
3b    V<Range> getRanges(Coord xl, Coord xr, Coord yl,
86                       Coord yr) {
9c      V<Range> res;
c3      rec(res,
33          lower_bound(xs.begin(), xs.end(), xl) - xs.begin(),
89          lower_bound(xs.begin(), xs.end(), xr) - xs.begin(),
9b          lower_bound(ys.begin(), ys.end(), yl) - ys.begin(),
48          lower_bound(ys.begin(), ys.end(), yr) - ys.begin(),
a9          0, n, lg);
26      return res;
3b    }
85  };
```

cartesian_tree.hpp

md5: 3171fc

```
4f  template <class T>
1e  vector<int> cartesian_tree(const vector<T>& a) {
55    int n = a.size(), t = 0;
89    vector<int> pa(n, -1), st(n);
1a    REP(i, n) {
13      int pr = -1;
3b      while (t && a[i] < a[st[t - 1]]) pr = st[--t];
58      if (pr != -1) pa[pr] = i;
25      if (t) pa[i] = st[t - 1];
59      st[t++] = i;
84    }
9a    return pa;
31  }
```

disjoint_table.hpp

md5: c59338

```
af  template <class S, S op(S, S), S e()>
12  struct DisjointTable {
4f    V<V<S>> a;
86    DisjointTable(V<S> v = {}) {
35      int lg = 1, n = 2;
b8      while (n < ssize(v)) lg++, n *= 2;
34      v.resize(n, e());
```

```
86     a.assign(lg, V<S>(n, e()));
d9     a[0] = v;
0b     REP(h, lg) if (h) {
84       int u = 1 << h;
ff       for (int m = u; m < n; m += 2 * u) {
d4         S q = e();
5c         REP(i, u)
28         a[h][m - i - 1] = q = op(v[m - i - 1], q);
02         q = e();
4a         REP(i, u) a[h][m + i] = q = op(q, v[m + i]);
0c       }
55     }
06   }
be   S query(int l, int r) {
39     r--;
80     if (l > r) return e();
40     if (l == r) return a[0][l];
a3     int u = 31 - countl_zero((unsigned int)(l ^ r));
03     return op(a[u][l], a[u][r]);
c2   }
c5 };
```

fastset.hpp

md5: 42fce0

```
66 struct FS {
40   using U = unsigned long long;
0e   const U B = 64;
21   U n;
fc   V<V<U>> a;
a4   FS(U n) : n(n) {
2a     do a.emplace_back(n = (n + B - 1) / B);
3b     while (n > 1);
6b   }
88   bool operator[](ll i) const {
5b     return a[0][i / B] >> (i % B) & 1;
be   }
6d   void insert(ll i) {
f9     for (auto& v : a) {
67       v[i / B] |= 1ULL << (i % B);
35       i /= B;
40     }
66   }
99   void erase(ll i) {
21     for (auto& v : a) {
bf       v[i / B] &= ~(1ULL << (i % B));
f3       if (v[i / B]) break;
81       i /= B;
cb     }
18   }
6a   ll next(ll i) {  // no less than i
c9     REP(h, a.size()) {
1c       if (i / B >= a[h].size()) break;
d1       U d = a[h][i / B] >> (i % B);
66       if (d) {
ad         i += countr_zero(d);
08         while (h--) i = i * B + countr_zero(a[h][i]);
6c         return i;
e6       }
18       i = i / B + 1;
c9     }
cd     return -1;
4f   }
aa   ll prev(ll i) {  // no greater than i
2a     REP(h, a.size()) {
62       if (i < 0) break;
1c       U d = a[h][i / B] << (~i % B);
93       if (d) {
27         i -= countl_zero(d);
4d         while (h--) i = i * B + __lg(a[h][i]);
80         return i;
8f       }
b9       i = i / B - 1;
80     }
48     return -1;
a3   }
42 };
```

fenwick.hpp

md5: 8cafd8

```
4f template <class T>
ff struct Fenwick {
c8   int n;
26   V<T> seg;
23   Fenwick(int n = 0) : n(n), seg(n + 1) {}
1b   void add(int i, T x) {
8c     i++;
c9     while (i <= n) {
96       seg[i] += x;
54       i += i & -i;
24     }
80   }
52   T sum(int i) {
77     T s = 0;
8c     while (i > 0) {
e4       s += seg[i];
18       i -= i & -i;
5c     }
bd     return s;
ad   }
74   T sum(int a, int b) { return sum(b) - sum(a); }
0f   int sum_lower_bound(T x) {
81     if (x <= seg[0]) return 0;
42     int res = 0, len = 1;
02     while (2 * len <= n) len *= 2;
8a     for (; len >= 1; len /= 2)
bf       if (res + len <= n && seg[res + len] < x)
65         x -= seg[res += len];
4d     return res + 1;
8c   }
8c };
```

kinetic_tournament_by_beats.hpp

md5: b4d8b4

```
72 struct Line {
81   ll a, b;
ad   ll cross(Line r) const {
9f     return r.a <= a ? INF : (b - r.b) / (r.a - a) + 1;
6d   }
27 };
c2 struct S {
3c   Line max;
f4   ll melt, fail = 0;
1d };
64 S e() { return {{0, -INF}, INF, 0}; }
96 S op(S l, S r) {
ac   S res;
ca   res.melt = min(l.melt, r.melt);
f3   bool fl = l.max.b < r.max.b;
da   res.max = fl ? r.max : l.max;
c7   if (fl) chmin(res.melt, r.max.cross(l.max));
de   if (!fl) chmin(res.melt, l.max.cross(r.max));
bb   return res;
e6 }
37 struct F {
7c   ll dt;
00 };
fe F id() { return {0}; }
ca S mapping(F f, S x) {
eb   x.melt -= f.dt;
c6   x.max.b += f.dt * x.max.a;
56   if (x.melt <= 0) x.fail = 1;
c8   return x;
55 }
da F composition(F f, F x) {
46   x.dt += f.dt;
66   return x;
b4 }
```

lazysegtree.hpp

md5: 9abb08

```
a8 template <class S, class F, S op(S, S),
36           F composition(F, F), S mapping(F, S)>
ee struct LazySegtree {
90   int n, s, log;
7a   S e;
e4   F id;
bf   V<S> d;
04   V<F> lz;

85   void update(int k) { d[k] = op(d[2 * k], d[2 * k + 1]); }
e7   void all_apply(int k, F f) {
78     d[k] = mapping(f, d[k]);
45     if (k < s) lz[k] = composition(f, lz[k]);
       // if(d[k].fail) push(k), update(k);
8f   }
3d   void push(int k) {
```

```
ae      all_apply(2 * k, lz[k]);
7c      all_apply(2 * k + 1, lz[k]);
ed      lz[k] = id;
74    }
ff    void pushx(int p) {
af      for (int i = log; i >= 1; i--) push(p >> i);
7b    }
1a    void updx(int p) {
d7      for (int i = 1; i <= log; i++) update(p >> i);
6c    }

6b    LazySegtree() = default;
ed    LazySegtree(unsigned n, S e, F id)
02        : n(n), s(bit_ceil(n)), e(e), id(id) {
cd      log = countr_zero((unsigned)s);
7b      d.assign(2 * s, e), lz.assign(s, id);
47    }
dc    LazySegtree(const vector<S>& v, S e, F id)
6c        : LazySegtree(v.size(), e, id) {
8f      ranges::copy(v, d.begin() + s);
78      for (int i = s - 1; i >= 1; i--) update(i);
2d    }
e4    void set(int p, S x) {
e6      pushx(p += s);
e3      d[p] = x;
7f      updx(p);
18    }
57    S get(int p) {
8f      pushx(p += s);
de      return d[p];
74    }
09    S prod(int l, int r) {
b5      if (l == r) return e;
ff      pushx(l += s), pushx((r += s) - 1);
ed      S sml = e, smr = e;
00      while (l < r) {
c8        if (l & 1) sml = op(sml, d[l++]);
41        if (r & 1) smr = op(d[--r], smr);
5f        l >>= 1, r >>= 1;
d0      }
32      return op(sml, smr);
02    }
fc    void apply(int l, int r, F f) {
b0      if (l == r) return;
03      if ((l += s) > s) pushx(l - 1);
cd      if ((r += s) < s * 2) pushx(r);
af      int l2 = l, r2 = r;
80      while (l < r) {
ba        if (l & 1) all_apply(l++, f);
36        if (r & 1) all_apply(--r, f);
bb        l >>= 1, r >>= 1;
78      }
64      if (l2 > s) updx(l2 - 1);
95      if (r2 < s * 2) updx(r2);
eb    }
58    template <class G>
35    int max_right(int l, G f) {
22      if (l == n) return n;
b5      pushx(l += s);
7f      S sm = e;
e5      while (1) {
03        while (l % 2 == 0) l >>= 1;
fd        S t = op(sm, d[l]);
86        if (!f(t)) break;
73        sm = t;
7a        if (l++, (l & -l) == l) return n;
35      }
6b      while (l < s) {
c4        push(l);
8d        l = 2 * l;
59        if (S t = op(sm, d[l]); f(t)) sm = t, l++;
de      }
30      return l - s;
e8    }
bc    template <class G>
79    int min_left(int r, G f) {
c8      if (r == 0) return 0;
c7      pushx((r += s) - 1);
a7      S sm = e;
99      while (1) {
d0        r--;
0f        while (r > 1 && (r % 2)) r >>= 1;
ad        S t = op(d[r], sm);
fc        if (!f(t)) break;
15        sm = t;
db        if ((r & -r) == r) return 0;
71      }
8f      while (r < s) {
e5        push(r);
91        r = 2 * r + 1;
7b        if (S t = op(d[r], sm); f(t)) sm = t, r--;
46      }
c3      return r + 1 - s;
d9    }
9a  };
```

li-chao-tree.hpp

md5: 25d839

```
95  template <class Func, class Eval>
50  struct LiChaoTreeFlexible {
e1    int n, N;
15    vector<Func> A;
e5    vector<bool> vis;
da    Eval ev;
f0    bool cmpat(Func fl, Func fr, int p) {
65      if (p >= n) p = n - 1;
b2      return ev(fl, p) < ev(fr, p);
41    }
91    LiChaoTreeFlexible(int n, Func inf, Eval eval)
1a        : n(n), N(bit_ceil(unsigned(n))), A(N * 2, inf),
71          vis(N * 2, 0), ev(move(eval)) {}
06    void addSegment(int l, int r, Func f) {
c2      if (l >= r) return;
29      auto dfs = [&](int i, Func f, int a, int b,
5a                     auto& dfs) -> void {
2f        vis[i] = 1;
93        if (i >= N * 2 || r <= a || b <= l) return;
e0        int m = (a + b) / 2;
f2        if (!(l <= a && b <= r)) {
1f          dfs(i * 2, f, a, m, dfs);
5a          dfs(i * 2 + 1, f, m, b, dfs);
36          return;
76        }
cd        if (cmpat(f, A[i], m)) swap(A[i], f);
20        if (a + 1 == b) return;
27        if (cmpat(f, A[i], a)) dfs(i * 2, f, a, m, dfs);
06        if (cmpat(f, A[i], b - 1))
15          dfs(i * 2 + 1, f, m, b, dfs);
55      };
8c      dfs(1, f, 0, N, dfs);
45    }
e2    void addLine(Func f) { addSegment(0, N, f); }
6f    Func minFunc(int p) {
f9      int i = 1, l = 0, r = N;
88      Func res = A[0];
60      while (i < N * 2 && vis[i]) {
b8        if (cmpat(A[i], res, p)) res = A[i];
ec        int m = (l + r) / 2;
2d        i = i * 2 + (m <= p);
42        (m <= p ? l : r) = m;
10      }
25      return res;
e9    }
25  };
```

pbds_rope.hpp

md5: d3e91e

```
77  #include <ext/pb_ds/assoc_container.hpp>
07  #include <ext/pb_ds/tree_policy.hpp>
99  using namespace __gnu_pbds;

    // std::setの上位互換
8b  tree<int, null_type, less<int>, rb_tree_tag,
d3       tree_order_statistics_node_update>
60    s;
99  s.find_by_order(k);  // k番目の要素のイテレータ
90  s.order_of_key(k);   // k未満が何個あるか

    // std::unordered_mapの上位互換(4倍くらい早いらしい)
    // 第3template引数にハッシュ関数指定可能
26  gp_hash_table<int, int> m;

    // ==============

75  #include <ext/rope>
e6  using namespace __gnu_cxx;
```

```
   // 計算量不明
27 rope<int> v0, v1, v2;
8a v0.insert(0, v1);  // 0番目にv1を挿入
95 v0.insert(0, 1);   // 0番目に1を挿入
32 v0.replace(0, 2, 1);  // 0番目から2要素を1つの1に変更
36 v0.replace(0, 2, v1);  // 0番目の2要素をv1に変更
2c v0.erase(0, 2);        // 0番目から2要素を削除
   // replaceやeraseは2つのイテレータで範囲指定もできる
ab v2 = v0.substr(0, 2);  // 0番目から2要素で1つのropeに
eb v0.mutable_begin(), v0.mutable_end();
   // 変更可能な両端イテレータ
d3 v0[2];  // 2番目の要素
   // resize, reserveはvectorと同じ
```

range_chmin_chmax_add_sum_beats.hpp md5: cd07a7

```
d4 struct F {
02   ll l, r, a;
b4 };
d6 template <class Cmp, ll INF>
83 struct ChminElem {
7a   ll x = INF, y = INF, c = 0;
67   ChminElem op(ChminElem r) const {
fa     if (x == r.x) return {x, min(y, r.y, Cmp()), c + r.c};
51     if (Cmp()(x, r.x)) return {x, min(y, r.x, Cmp()), c};
c4     return {r.x, min(x, r.y, Cmp()), r.c};
d4   }
43   void apply(F f) {
fc     if (x != INF) x = max(f.l, min(f.r, x)) + f.a;
9f     if (y != INF) y = max(f.l, min(f.r, y)) + f.a;
73   }
16 };
c6 struct Elem {
b9   ChminElem<less<ll>, INF> l;
12   ChminElem<greater<ll>, -INF> r;
48   ll cnt = 0, sum = 0;
5f   ll fail = 0;
f1 };
1d Elem op(Elem l, Elem r) {
0a   return {l.l.op(r.l), l.r.op(r.r), l.cnt + r.cnt,
94           l.sum + r.sum, 0};
31 }
f0 F composition(F f, F x) {
31   if (f.l != -INF) {
da     f.l -= x.a;
65     chmax(x.l, f.l);
20     chmax(x.r, f.l);
48   }
f8   if (f.r != INF) {
8c     f.r -= x.a;
26     chmin(x.l, f.r);
33     chmin(x.r, f.r);
f2   }
58   x.a += f.a;
04   return x;
bd }
f0 Elem mapping(F f, Elem x) {
2d   x.fail = 1;
90   if (f.l != -INF && x.l.y <= f.l) return x;
8a   if (f.r != INF && x.r.y >= f.r) return x;
79   if (x.l.x < f.l) x.sum += (f.l - x.l.x) * x.l.c;
3b   if (x.r.x > f.r) x.sum += (f.r - x.r.x) * x.r.c;
b1   x.sum += f.a * x.cnt;
b5   x.l.apply(f);
52   x.r.apply(f);
5a   x.fail = 0;
c3   return x;
34 }
2f Elem e() { return Elem(); }
cd F id() { return {-INF, INF, 0}; }
```

segtree.hpp md5: 5cd8d9

```
74 template <class S, S op(S, S)>
6c struct Segtree {
16   int n, s;
5d   S e;
a0   V<S> d;
9d   Segtree() = default;
2b   Segtree(unsigned n, S e) : n(n), s(bit_ceil(n)), e(e) {
67     d.assign(2 * s, e);
67   }
d9   Segtree(const vector<S>& v, S e) : Segtree(v.size(), e) {
f5     ranges::copy(v, d.begin() + s);
c2     for (int i = s - 1; i >= 1; i--)
20       d[i] = op(d[2 * i], d[2 * i + 1]);
cc   }
37   void set(int p, S x) {
3e     d[p += s] = x;
28     while ((p >>= 1) >= 1)
52       d[p] = op(d[2 * p], d[2 * p + 1]);
b3   }
c3   S get(int p) { return d[p + s]; }
ad   S prod(int l, int r) {
19     S sml = e, smr = e;
c3     l += s, r += s;
40     while (l < r) {
e7       if (l & 1) sml = op(sml, d[l++]);
5c       if (r & 1) smr = op(d[--r], smr);
62       l >>= 1, r >>= 1;
30     }
8e     return op(sml, smr);
d3   }
26   template <class G>
fd   int max_right(int l, G f) {
f8     if (l == n) return n;
34     l += s;
15     S sm = e;
e4     while (1) {
80       while (l % 2 == 0) l >>= 1;
40       S t = op(sm, d[l]);
1e       if (!f(t)) break;
d2       sm = t;
a0       if (l++, (l & -l) == l) return n;
f0     }
e7     while (l < s) {
3d       l = 2 * l;
db       if (S t = op(sm, d[l]); f(t)) sm = t, l++;
25     }
6d     return l - s;
b4   }
c3   template <class F>
6b   int min_left(int r, F f) {
13     if (r == 0) return 0;
1f     r += s;
0f     S sm = e;
14     while (1) {
48       r--;
cb       while (r > 1 && (r % 2)) r >>= 1;
8a       S t = op(d[r], sm);
e2       if (!f(t)) break;
e1       sm = t;
9f       if ((r & -r) == r) return 0;
4f     }
c2     while (r < s) {
99       r = 2 * r + 1;
21       if (S t = op(d[r], sm); f(t)) sm = t, r--;
4e     }
d2     return r + 1 - s;
9a   }
5c };
```

unionfind.hpp md5: 5a6135

```
0c struct UnionFind {
76   V<int> p;
df   int gn;
6c   UnionFind(int n = 0) : p(n, -1), gn(n) {}
c4   void merge(int a, int b) {
0b     int x = group(a), y = group(b);
bb     if (x == y) return;
e4     gn--;
74     if (-p[x] < -p[y]) swap(x, y);
ac     p[x] += p[y];
43     p[y] = x;
fc   }
75   int size(int a) { return -p[group(a)]; }
f1   int group(int a) {
a3     if (p[a] < 0) return a;
1a     return p[a] = group(p[a]);
bf   }
b9   bool same(int a, int b) { return group(a) == group(b); }
5a };
```

wavelet_matrix.hpp

md5: 9e581a

```
09  struct BitVector {
c4    ll n, b, m;
9e    V<unsigned> bit, sum;

75    BitVector(ll n = 0)
77        : n(n), b(n / 32 + 1), bit(b), sum(b + 1) {}
0e    void build() {
80      REP(i, b) sum[i + 1] = sum[i] + popcount(bit[i]);
6d      m = n - sum[b];
47    }
13    int rank(int k) {
51      return sum[k >> 5] +
c6             popcount(bit[k >> 5] & ~(~0u << (k & 31)));
10    }
12    int rank(bool val, int k) {
de      return val ? rank(k) : k - rank(k);
92    }
c7  };

34  template <typename T, int MAXLOG>
29  struct WaveletMatrix {
a4    size_t n;
7d    BitVector a[MAXLOG];
d0    WaveletMatrix(V<T> v = {}) : n(v.size()) {
07      V<T> l(n), r(n);
8c      for (int d = MAXLOG - 1; d >= 0; d--) {
78        a[d] = BitVector(n);
20        int x = 0, y = 0;
81        REP(i, n) {
a5          unsigned h = (v[i] >> d) & 1;
78          a[d].bit[i >> 5] |= h << (i & 31);
57          (h ? r[y++] : l[x++]) = v[i];
ac        }
4f        a[d].build();
02        v.swap(l);
91        REP(i, y) v[x + i] = r[i];
89      }
27    }
ba    pair<int, int> succ(bool f, int l, int r, int d) {
5f      return {a[d].rank(f, l) + a[d].m * f,
2a              a[d].rank(f, r) + a[d].m * f};
f8    }
      // count i s.t. (0 <= i < r) && v[i] == x
e7    int rank(const T &x, int r) {
f4      int l = 0;
3e      for (int d = MAXLOG - 1; d >= 0; d--) {
8a        tie(l, r) = succ((x >> d) & 1, l, r, d);
d3      }
97      return r - l;
51    }
      // k-th(0-indexed) smallest number in v[l,r)
71    T kth_smallest(int l, int r, int k) {
f5      assert(0 <= k && k < r - l);
92      T ret = 0;
3c      for (int d = MAXLOG - 1; d >= 0; d--) {
b4        int cnt = a[d].rank(0, r) - a[d].rank(0, l);
a9        ll f = cnt <= k;
cc        ret |= T(f) << d;
35        k -= cnt * f;
68        tie(l, r) = succ(f, l, r, d);
1c      }
63      return ret;
d5    }
      // k-th(0-indexed) largest number in v[l,r)
a7    T kth_largest(int l, int r, int k) {
26      return kth_smallest(l, r, r - l - k - 1);
07    }
      // count i s.t. (l <= i < r) && (v[i] < upper)
1e    int range_freq(int l, int r, T upper) {
59      int ret = 0;
1d      for (int d = MAXLOG - 1; d >= 0; d--) {
03        bool f = (upper >> d) & 1;
3c        if (f) ret += a[d].rank(0, r) - a[d].rank(0, l);
3d        tie(l, r) = succ(f, l, r, d);
2c      }
a4      return ret;
24    }
      // count i s.t. (l <= i < r) && (lower <= v[i] < upper)
16    int range_freq(int l, int r, T lower, T upper) {
5c      return range_freq(l, r, upper) -
0b             range_freq(l, r, lower);
81    }
      // max v[i] s.t. (l <= i < r) && (v[i] < upper)
c1    T prev_value(int l, int r, T upper) {
7a      int cnt = range_freq(l, r, upper);
94      return cnt == 0 ? T(-1) : kth_smallest(l, r, cnt - 1);
08    }
      // min v[i] s.t. (l <= i < r) && (lower <= v[i])
60    T next_value(int l, int r, T lower) {
ce      int cnt = range_freq(l, r, lower);
2e      return cnt == r - l ? T(-1) : kth_smallest(l, r, cnt);
fc    }
9e  };
```

number-theory

bsgs.hpp

md5: 8c2ceb

$x^e s = t$ を満たす n 未満の非負整数 e が存在すればその最小値を返す(なければ-1)

- op: 作用素の積
- mapping: 作用

op を $O(\log m)$ 回、mapping を $O(m + n/m)$ 回呼ぶ

```
da  template <class X, class S>
e6  ll bsgs(X x, S s, S t, auto op, X xe, auto mapping, ll n) {
92    ll m = sqrt(n), k = 0;
2c    S xit = t, p = s, nxp = s;
81    X x2 = x, xm = x;
56    unordered_set<S> mp;
e3    REP(i, m) mp.insert(xit = mapping(x, xit));
8e    for (ll d = 1; d < m; d *= 2) {
27      if ((m - 1) & d) xm = op(xm, x2);
21      x2 = op(x2, x2);
00    }
a1    REP(a, 2) {
73      while (!mp.contains(nxp = mapping(xm, p))) {
02        if (k >= n) return -1;
44        p = nxp;
f1        k += m;
d6      }
ee      REP(i, m) {
4b        if (p == t) return k;
31        p = mapping(x, p);
97        k++;
96      }
09    }
6e    return -1;
8c  }
```

crt.hpp

md5: 590ea0

- ext_gcd: $ax + by = g = \gcd(a, b)$
- crt: mod は互いに素でなくて良い。 (rem, mod) を与えると (rem, mod) を返す。

```
3f  struct EG {
c4    ll g, x, y;
81  };
2e  EG ext_gcd(ll a, ll b) {
3e    if (!b) return a >= 0 ? EG{a, 1, 0} : EG{-a, -1, 0};
15    auto e = ext_gcd(b, a % b);
5f    return EG{e.g, e.y, e.x - a / b * e.y};
04  }
1e  ll inv_mod(ll x, ll md) {
57    return (ext_gcd(x, md).x % md + md) % md;
e5  }
6b  template <class T, class U>
ad  T pow_mod(T x, U n, T md) {
01    T r = 1 % md;
85    x %= md;
d7    while (n) {
29      if (n & 1) r = (r * x) % md;
e2      x = (x * x) % md;
bf      n >>= 1;
d8    }
b3    return r;
ae  }
5e  pair<ll, ll> crt(const V<ll>& b, const V<ll>& c) {
fb    ll r = 0, m = 1;
```

```
19    REP(i, b.size()) {
87      auto eg = ext_gcd(m, c[i]);
81      ll g = eg.g, im = eg.x;
73      if ((b[i] - r) % g) return {0, -1};
b2      ll tmp = (b[i] - r) / g * im % (c[i] / g);
7f      r += m * tmp;
61      m *= c[i] / g;
30    }
34    return {(r % m + m) % m, m};
59  }
```

floor-sum.hpp

md5: 9ab5a1

- $\sum_{i=0}^{n-1} \lfloor \frac{a \times i + b}{m} \rfloor$
- $a, m > 0, b \geq 0$

```
fd  ll gauss_sum(ll n, ll a, ll b, ll m) {
61    ll res = 0;
ad    while (n) {
9e      res += a / m * (n * (n - 1) / 2);
ea      a %= m;
d8      res += b / m * n;
b3      b %= m;
85      ll y = a * n + b;
99      n = y / m;
89      b = y % m;
8f      y = a, a = m, m = y;
74    }
92    return res;
9a  }
```

floor_sum_monoid.hpp

md5: cff8aa

```
    // 1 <= A, 0 <= B
bb  template <class Int, class S, S op(const S&, const S&)>
f4  S FloorSumMonoid(Int N, Int M, Int A, Int B, S X, S Y,
9b                   S e) {
0f    auto pow = [&](S r, S a, Int i) {
23      do {
2c        if (i % 2) r = op(r, a);
e6        a = op(a, a);
c0      } while (i /= 2);
40      return r;
dd    };
e5    S L = e, R = e;
e0    Int C = (A * N + B) / M;
85    while (1) {
18      Int qa = A / M, qb = B / M;
f2      A %= M;
6c      B %= M;
ca      X = pow(X, Y, qa);
12      L = pow(L, Y, qb);
97      C -= qa * N + qb;
cb      if (C == 0) break;
5c      Int D = (M * C - B - 1) / A + 1;
32      R = op(pow(Y, X, N - D), R);
39      B = M - B - 1 + A;
45      N = C - 1;
ec      C = D;
d1      swap(M, A);
0e      swap(X, Y);
ff    }
82    return op(pow(L, X, N), R);
cf  }
    // struct S { ll x, y, sum; };
    // S op(const S& l, const S& r){
    //   return { l.x + r.x, l.y + r.y, l.sum + r.sum + l.y *
    //   r.x };
    // }
    // S e = { 0,0,0 };
```

garner.hpp

md5: 3be2e2

- $O(N^2)$. modは互いに素である必要がある
    - そうでない場合は各ペアについてcoprimize...を呼ぶ

```
    // crt.hpp の inv_mod を使用
    // 1 回実行 : ll ans = garner<ll>(md)(rem);
04  template <class E>
d9  struct garner {
36    int n;
e5    V<ll> m, h;
92    garner(V<ll> md) : n(md.size()), m(md), h(n, 1) {
2c      REP(i, n) REP(j, i) h[i] = h[i] * m[j] % m[i];
ff      REP(i, n) h[i] = inv_mod(h[i], m[i]);
84    }
2b    E operator()(const V<ll>& r) {
c2      V<ll> a(n, 1), b(n, 0);
29      E resa = 1, res = 0;
44      REP(k, n) {
06        ll t = (r[k] - b[k]) * h[k] % m[k];
30        if (t < 0) t += m[k];
6c        for (int i = k + 1; i < n; ++i) {
8b          b[i] = (b[i] + t * a[i]) % m[i];
dd          a[i] = a[i] * m[k] % m[i];
da        }
63        res = res + E(t) * resa;
2f        resa = resa * E(m[k]);
cf      }
d9      return res;
9f    }
13  };
    // 連立合同式の mod が互いに素になるように変形
    // 返り値 : 解ありなら 1 、解なしなら 0
82  bool coprimize_mod_system(ll& r1, ll& m1, ll& r2, ll& m2) {
45    ll g = gcd(m1, m2);
98    if ((r2 - r1) % g) return 0;
e0    g = gcd(m2 / g, m1);
85    while (g > 1) m1 /= g = gcd(g, m1);
48    m2 /= gcd(m1, m2);
03    r1 %= m1, r2 %= m2;
46    return 1;
3b  }
```

miller_rabin.hpp

md5: e031e6

```
0a  bool is_prime(ll n) {
38    if (n <= 1) return 0;
2d    if (n % 2 == 0) return n == 2;
f0    ll d = n - 1;
b1    while (d % 2 == 0) d /= 2;
      // n < 4.7e9 なら {2, 7, 61}
      // n < 3.4e14なら{17以下の素数} で十分
72    for (ll a :
fb         {2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37}) {
98      if (n <= a) break;
fd      ll t = d, y = pow_mod<__int128_t>(a, t, n);  // over
82      for (; t != n - 1 && y != 1 && y != n - 1; t *= 2)
e9        y = __int128_t(y) * y % n;  // flow
78      if (y != n - 1 && t % 2 == 0) return 0;
33    }
54    return 1;
09  }
a8  ll pollard_single(ll n) {
0a    if (n % 2 == 0) return 2;
39    REP(st, n) if (st) {
e1      auto f = [&](ll x) {
2d        return (__int128_t(x) * x + st) % n;
6d      };
52      ll x = st, y = f(x), p;
89      while ((p = gcd((y - x + n), n)) != n) {
17        if (p != 1) return p;
85        x = f(x), y = f(f(y));
e1      }
fb    }
00  }
41  V<ll> pollard(ll n) {
cd    if (n == 1) return {};
89    if (is_prime(n)) return {n};
1a    ll x = pollard_single(n);
54    auto res = pollard(x);
79    for (auto a : pollard(n / x)) res.push_back(a);
b4    return res;
e0  }
```

modint.hpp

md5: 71bc28

```
92  template <int MD, int g = 3>
31  struct ModInt {
a8    using M = ModInt;
9e    const static inline M G = g;
c0    unsigned int v;
06    ModInt() : v(0) {}
```

```
8a    ModInt(ll w) : v(w % MD + MD) {
d7      if (v >= MD) v -= MD;
a3    }
7c    static M raw(unsigned int v) {
75      M res;
37      res.v = (v < MD) ? v : v - MD;
ce      return res;
41    }
d4    explicit operator bool() const { return v != 0; }
ce    M operator-() const { return M() - *this; }
5e    M operator+(M r) const { return raw(v + r.v); }
f5    M operator-(M r) const { return raw(v + MD - r.v); }
5f    M operator*(M r) const { return raw(ll(v) * r.v % MD); }
2d    M operator/(M r) const { return *this * r.inv(); }
8e    M& operator+=(M r) { return *this = *this + r; }
fb    M& operator-=(M r) { return *this = *this - r; }
f2    M& operator*=(M r) { return *this = *this * r; }
16    M& operator/=(M r) { return *this = *this / r; }
0f    bool operator==(M r) const { return v == r.v; }
ff    M pow(ll n) const {
1d      M x = *this, r = 1;
e1      while (n) {
d3        if (n & 1) r *= x;
00        x *= x;
59        n >>= 1;
2a      }
c8      return r;
a8    }
a2    M inv() const { return pow(MD - 2); }
31    friend ostream& operator<<(ostream& os, M r) {
83      return os << r.v;
c6    }
71  };
    // using Mint = ModInt<998244353, 3>;
```

tonelli_shanks.hpp

md5: fd8d35

```
    // x^2=N(mod p : prime) となるxを返す(存在しないなら-1)
    // pow_mod(a,b,p)はa^b(mod p)
d7  ll Tonelli_Shanks(ll N, ll p) {
d9    N %= p;
de    if (p == 2) return N;
c9    if (pow_mod(N, p / 2, p) == p - 1) return -1;
3e    if (p % 4 == 3) return pow_mod(N, (p + 1) / 4, p);
c7    ll n = 1, pp = p - 1, c = 0;
ad    while (pow_mod(n, p / 2, p) != p - 1) n++;
1b    while (pp % 2 == 0) pp /= 2, c++;
d2    ll s = pow_mod(N, pp, p);
2f    ll r = pow_mod(N, (pp + 1) / 2, p);
2f    for (int i = c - 2; i >= 0; --i) {
53      if (pow_mod(s, 1LL << i, p) == p - 1) {
0c        s = s * pow_mod(n, p >> (1 + i), p) % p;
a5        r = r * pow_mod(n, p >> (2 + i), p) % p;
4c      }
9b    }
7a    return r;
fd  }
```

fps

multieval.hpp

md5: 5fd568

```
71  template <class D>
e7  struct MultiEval {
18    int n;
33    V<V<D>> H;
7f    MultiEval(V<D> p) : n(p.size()), H(n * 2) {
d5      REP(i, n) H[n + i] = {1, -p[i]};
aa      for (int i = n - 1; i >= 1; i--)
98        H[i] = multiply(H[i * 2], H[i * 2 + 1]);
cb    }
86    V<D> inv(V<D> f, int m) {
77      f.resize(m * 2);
e0      V<D> q = {1};
e1      for (int z = 2; z < m * 2; z *= 2) {
bb        auto g = multiply(q, q);
2e        g = multiply(g, V<D>(f.begin(), f.begin() + z));
48        q.resize(z);
0e        REP(i, z) q[i] += q[i] - g[i];
97      }
b3      q.resize(m);
a1      return q;
07    }
66    V<D> query(V<D> pol) {
1d      V<V<D>> G(n * 2);
6e      auto f = inv(H[1], pol.size());
9a      pol.resize(pol.size() + n - 1);
3e      G[1] = tr_multiply(f, pol);
a7      for (int i = 2; i < n * 2; i++)
7e        G[i] = tr_multiply(H[i ^ 1], G[i / 2]);
81      REP(i, n) G[1][i] = G[n + i][0];
1a      return G[1];
78    }
5f  };
```

bostan_mori.hpp

md5: cbe512

```
    // res[i] = 1/d の N-i 次の係数
71  template <class D>
1e  V<D> bostan_mori_sub(V<D> d, ll N) {
f1    int k = d.size();
8c    if (!N) {
e7      V<D> res(k);
80      res[0] = d[0].inv();
3a      return res;
d1    }
54    V<D> nd(k + 1), q(k * 2);
88    REP(i, k) nd[i] = i % 2 ? -d[i] : d[i];
b6    auto h = multiply(d, nd);
bf    REP(i, k) h[i] = h[i * 2];
a8    h.resize(k);
0e    h = bostan_mori_sub(move(h), N / 2);
94    REP(i, k) q[i * 2 + N % 2] = h[i];
4e    return tr_multiply(move(nd), move(q));
c4  }
    // res[i] = n/d の N+i 次の係数
77  template <class D>
78  V<D> bostan_mori(V<D> n, V<D> d, ll N, int m) {
6a    auto inv = [](V<D> f, int m) {
e3      f.resize(m * 2);
5e      V<D> q = {f[0].inv()};
ff      for (int z = 2; z / 2 < m; z *= 2) {
04        auto g = multiply(q, q);
ec        g = multiply(g, V<D>(f.begin(), f.begin() + z));
d5        q.resize(z);
49        REP(i, z) q[i] += q[i] - g[i];
2b      }
46      q.resize(m);
dd      return q;
d0    };
bf    int k = max(n.size(), d.size());
2a    n.resize(k);
48    d.resize(k);
00    auto f = bostan_mori_sub(d, N);
5f    reverse(f.begin(), f.end());
25    reverse(n.begin(), n.end());
d6    f = multiply(f, d);
10    f.resize(k);
a0    f = multiply(f, inv(d, k + m - 1));
6a    f.resize(k + m - 1);
f4    return tr_multiply(n, f);
cb  }
```

Interpolation.hpp

md5: 55b5f1

```
    // i=0,1,...,N-1についてf(a+i*b)=Y[i]となるとき、f(x)を返す
e7  Mint Interpolation_AP(int N, Mint a, Mint b, V<Mint> Y,
2e                        Mint x) {
0c    Mint f = 1;
5b    REP(i, N) if (i) f *= b * i;
bd    f = f.inv();  // 逆元
b8    V<Mint> C = Y, L(N + 1, 1), R(N + 1, 1);
da    for (int i = N - 2; i >= 0; i -= 2) C[i] = -C[i];
35    for (int i = N - 1; i >= 0; --i) {
79      C[i] *= f;
9a      C[N - 1 - i] *= f;
ff      f *= b * i;
ce    }
40    REP(i, N) {
63      L[i + 1] = L[i] * (x - (a + b * i));
75      R[N - 1 - i] = R[N - i] * (x - (a + b * (N - 1 - i)));
59    }
03    Mint s = 0;
f4    REP(i, N) s += L[i] * R[i + 1] * C[i];
bb    return s;
```

```
9d  }

    // i=0,1,...,N-1についてf(X[i])=Y[i]となるとき、f(x)を返す
4c  Mint Interpolation(int N, V<Mint> X, V<Mint> Y, Mint x) {
82    V<Mint> C(N, 1), L(N + 1, 1), R(N + 1, 1);
0f    REP(i, N) REP(j, N) if (i != j) C[i] *= X[i] - X[j];
e3    REP(i, N) C[i] = C[i].inv() * Y[i];  // 逆元
5a    REP(i, N) {
b9      L[i + 1] = L[i] * (x - X[i]);
1b      R[N - 1 - i] = R[N - i] * (x - X[N - 1 - i]);
22    }
19    Mint s = 0;
fc    REP(i, N) s += L[i] * R[i + 1] * C[i];
4e    return s;
55  }
```

berlekamp_massey.hpp

md5: 0c1df1

```
    // 有理母関数の分母を返す
5d  template <class Mint>
bf  vector<Mint> berlekamp_massey(const vector<Mint>& s) {
86    int n = int(s.size());
13    vector<Mint> b = {1}, c = {1}, t;
0a    Mint y = 1;
34    REP(ed, n) {
1f      int l = c.size(), m = b.size() + 1;
93      Mint x = 0;
df      REP(i, l) x -= c[i] * s[ed - i];
69      b.insert(begin(b), 0);
cb      if (!x.v) continue;
a4      Mint q = x / y;
d6      if (l < m) t = c, c.resize(m, 0);
a3      REP(i, m) c[i] -= q * b[i];
c9      if (l < m) b = t, y = x;
f1    }
a8    return c;
0c  }
```

fps.hpp

md5: d8237a

```
71  template <class D>
e7  struct Poly {
93    using P = const Poly&;
15    V<D> v;
3a    Poly(V<D> v = {}) : v(move(v)) {}
e9    void shrink() {
57      while (v.size() && !v.back()) v.pop_back();
83    }
6c    int size() const { return v.size(); }
30    D freq(int p) const {
65      return 0 <= p && p < size() ? v[p] : D(0);
ad    }
3b    Poly operator+(P r) const {
3e      V<D> res(max(size(), r.size()));
aa      REP(i, res.size()) res[i] = freq(i) + r.freq(i);
34      return res;
99    }
65    Poly operator-(P r) const {
d8      V<D> res(max(size(), r.size()));
d2      REP(i, res.size()) res[i] = freq(i) - r.freq(i);
e9      return res;
67    }
ea    Poly operator*(P r) const { return {multiply(v, r.v)}; }
23    Poly operator*(D r) const {
6e      V<D> res = v;
3c      REP(i, size()) res[i] *= r;
36      return res;
55    }
77    Poly operator/(D r) const { return *this * r.inv(); }
79    Poly operator/(Poly r) const {
19      if (size() < r.size()) return {{}};
c0      r.shrink();
52      int n = size() - r.size() + 1;
8a      return (rev().pre(n) * r.rev().inv(n)).rev(n);  // 変更
b3    }
66    Poly operator%(P r) const {
57      auto x = *this - *this / r * r;
ce      x.shrink();
f4      return x;
dc    }
45    Poly& operator+=(P r) { return *this = *this + r; }
92    Poly& operator-=(P r) { return *this = *this - r; }
d3    Poly& operator*=(P r) { return *this = *this * r; }
15    Poly& operator*=(D r) { return *this = *this * r; }
02    Poly& operator/=(P r) { return *this = *this / r; }
f3    Poly& operator/=(D r) { return *this = *this / r; }
73    Poly& operator%=(P r) { return *this = *this % r; }
a7    Poly clip(int l = 0, int r = -(1 << 30)) const {
64      if (r == -(1 << 30)) r = size();
5a      if (r < l) r = l;
14      V<D> res(r - l);
08      REP(i, r - l) res[i] = freq(i + l);
f1      return res;
51    }
81    Poly pre(int le) const { return clip(0, le); }
25    Poly rev(int n = -1) const {
54      auto res = v;
9c      if (n != -1) res.resize(n);
4f      reverse(res.begin(), res.end());
d4      return res;
ae    }
cb    Poly diff() const {
29      auto res = clip(1).v;
a8      REP(i, res.size()) res[i] *= i + 1;
d0      return res;
64    }
68    Poly inte() const {
ba      auto res = clip(-1).v;
0c      REP(i, res.size()) if (i) res[i] /= i;
23      return res;
1f    }
      // f * f.inv() = 1 + g(x)x^m
65    Poly inv(int m) const {
e1      Poly f = Poly({D(1) / freq(0)});
b0      for (int i = 1; i < m; i *= 2)
3b        f = (f * D(2) - f * f * pre(2 * i)).pre(2 * i);
12      return f.pre(m);
ea    }
d6    Poly exp(int n) const {
91      assert(freq(0) == 0);
7b      Poly f({1}), g({1}), q;
84      for (int i = 2; i < n * 2; i *= 2) {
9e        g = (g * 2 - f * g * g).pre(i / 2);
46        q = pre(i / 2).diff();
a7        q += (g * (f.diff() - f * q)).pre(i - 1);
03        f += (f * (pre(i) - q.inte())).pre(i);
6a      }
67      return f.pre(n);
15    }
ad    Poly log(int n) const {
29      assert(freq(0) == 1);
39      auto f = pre(n);
58      return (f.diff() * f.inv(n - 1)).pre(n - 1).inte();
b1    }
36    friend ostream& operator<<(ostream& os, P p) {
e9      if (p.size() == 0) return os << "0";
cc      for (auto i = 0; i < p.size(); i++) {
c9        if (p.v[i]) {
b3          os << p.v[i] << "x^" << i;
36          if (i != p.size() - 1) os << "+";
fb        }
fb      }
b1      return os;
38    }
d8  };
```

fps_composition.hpp

md5: 9c217c

```
    // f(g(x)) mod x^N (g[0] = 0)
71  template <class D>
f1  V<D> fps_composition(V<D> f, V<D> g, ll N) {
b8    ll z = 1;
06    while (z < N) z *= 2;
4e    V<D> p(z * 2), q(z * 3), res(N);
6d    REP(i, f.size()) p[i * 2] = f[i];
b8    REP(i, min<ll>(g.size(), z)) q[z * 2 + i] = -g[i];
dd    q[0] = 1;
84    auto dfs = [&](auto& h, V<D> q, ll u, ll v) -> void {
e8      if (v == z) return;
2c      ll nu = (u + 1) / 2, nv = v * 2;
79      V<D> nq = q, y(nu * 2 * (nv + 1));
81      REP(i, q.size()) if (i % 2) nq[i] = -nq[i];
62      q = multiply(q, nq);
1c      REP(d, nv + 1)
4d      REP(i, nu) y[d * nu * 2 + i] = q[d * u * 2 + i * 2];
89      h(h, move(y), nu, nv);
```

```
92      V<D> x(u * 2 * (v * 2 + 1) - 1);
29      REP(d, nv)
bd      REP(i, nu) x[d * u * 2 + i * 2 + 1] =
45          p[d * nu * 2 + i];
7a      p = tr_multiply(nq, x);
e4    };
8f    dfs(dfs, move(q), z, 1);
8c    REP(i, N) res[i] = p[z - 1 - i];
da    return res;
9c  }
```

fps_power_projection.hpp

md5: c000a5

```
    // [x^K] f(x) / (1 - y g(x)) mod y^N (g[0] = 0)
71  template <class D>
0a  V<D> power_projection(V<D> f, V<D> g, ll K, ll N) {
14    ll u = K + 1, v = 1;
12    V<D> p(u * 2 * v), q(u * 2 * (v + 1)), res(N);
8e    REP(i, min<ll>(f.size(), u)) p[i] = f[i];
f4    REP(i, min<ll>(g.size(), u)) q[u * 2 + i] = -g[i];
4f    q[0] = 1;
7a    while (K || v < N) {
03      ll nu = (u + 1) / 2, nv = v * 2;
e2      V<D> nq = q, x(nu * 2 * nv), y(nu * 2 * (nv + 1));
89      REP(d, v + 1)
29      REP(i, u) if (i % 2) nq[d * u * 2 + i] *= -1;
e7      q = multiply(q, nq);
21      p = multiply(p, nq);
19      REP(d, nv)
db      REP(i, nu) x[d * nu * 2 + i] =
6d          p[d * u * 2 + i * 2 + K % 2];
88      REP(d, nv + 1)
cb      REP(i, nu) y[d * nu * 2 + i] = q[d * u * 2 + i * 2];
7a      u = nu, v = nv, K /= 2;
2a      swap(p, x);
3e      swap(q, y);
8b    }
c8    REP(i, N) res[i] = p[i * 2];
ac    return res;
c0  }
```

garner_convolution.hpp

md5: eaba62

```
    // number-theory/modint.hpp , 同/crt.hpp (inv_mod まで) ,
    // 同/garner.hpp , fps/nft.hpp
cb  template <int MOD, int g>
78  V<int> multiply_mod(const V<int>& a, const V<int>& b) {
d4    V<ModInt<MOD, g>> c, d;
f8    for (auto x : a) c.push_back(x);
7b    for (auto x : b) d.push_back(x);
bd    c = multiply(c, d);
7f    V<int> res;
a9    for (auto x : c) res.push_back(x.v);
1b    return res;
fa  }
b7  template <class mint>
ef  V<mint> multiply_garner(V<mint> a, V<mint> b) {
1a    constexpr int m[] = {167772161, 469762049, 754974721};
00    auto g = garner<mint>(V<ll>(m, m + 3));
56    V<int> c(a.size()), d(b.size());
89    REP(i, a.size()) c[i] = a[i].v;
bf    REP(i, b.size()) d[i] = b[i].v;
41    V<V<int>> res(3);
da    res[0] = multiply_mod<m[0], 3>(c, d);
de    res[1] = multiply_mod<m[1], 3>(c, d);
c3    res[2] = multiply_mod<m[2], 11>(c, d);
a6    V<ll> r(3);
72    V<mint> ans(res[0].size());
23    REP(i, res[0].size()) {
70      REP(j, 3) r[j] = res[j][i];
e3      ans[i] = g(r);
08    }
e2    return ans;
ea  }
```

nft.hpp

md5: 4d52c9

```
    // 167772161     3      5 * 2^25 + 1
    // 377487361     7     45 * 2^23 + 1
    // 469762049     3      7 * 2^26 + 1
    // 595591169     3     71 * 2^23 + 1
    // 645922817     3     77 * 2^23 + 1
    // 754974721    11     45 * 2^24 + 1
    // 880803841    26    105 * 2^23 + 1
    // 897581057     3    107 * 2^23 + 1
    // 998244353     3    119 * 2^23 + 1
5d  template <class Mint>
87  void nft(bool type, V<Mint>& a) {
f2    int n = int(a.size()), s = 0;
31    while ((1 << s) < n) s++;
19    assert(1 << s == n);
cd    static V<Mint> ep;
b5    while (int(ep.size()) <= s) {
dd      ep.push_back(Mint::G.pow(Mint(-1).v >> ep.size()));
b5    }
cc    V<Mint> b(n);
d1    REP(i, s) {
12      int w = 1 << (s - i - 1);
c6      Mint now = 1;
22      for (int y = 0; y < n / 2; y += w) {
7a        REP(x, w) {
7d          auto l = a[y << 1 | x];
7a          auto r = now * a[y << 1 | x | w];
56          b[y | x] = l + r;
e0          b[y | x | n >> 1] = l - r;
1d        }
01        now *= ep[i + 1];
c9      }
13      swap(a, b);
b4    }
cc    if (type) reverse(a.begin() + 1, a.end());
1a  }
80  template <class Mint>
ac  V<Mint> cyconv(V<Mint> a, V<Mint> b, int k) {
75    int z = 1;
60    while (z < k) z *= 2;
c1    a.resize(z);
ec    b.resize(z);
8b    nft(0, a);
17    nft(0, b);
b7    Mint iz = Mint(z).inv();
4c    REP(i, z) a[i] *= b[i] * iz;
f7    nft(1, a);
14    a.resize(k);
39    return a;
2e  }
1e  template <class Mint>
1c  V<Mint> multiply(V<Mint> a, V<Mint> b) {
df    int n = a.size(), m = b.size();
70    if (!n || !m) return {};
ad    if (min(n, m) <= 32) {
b7      V<Mint> ans(n + m - 1);
df      REP(i, n) REP(j, m) ans[i + j] += a[i] * b[j];
63      return ans;
03    }
2f    return cyconv(move(a), move(b), n + m - 1);
44  }
    // multieval, bostan_mori, composition
4e  template <class Mint>
02  V<Mint> tr_multiply(V<Mint> a, V<Mint> b) {
ae    int n = a.size(), m = b.size();
e0    reverse(b.begin(), b.end());
19    b = cyconv(move(a), move(b), m);
f0    reverse(b.begin(), b.end());
17    b.resize(m - n + 1);
83    return b;
4d  }
```

relaxed_convolution.hpp

md5: ed3c47

```
    // f(i,c)はc=C[i-1]の時にA[i],B[i]を返す関数 (i=0の時はcは何でも
よい)
71  template <class D>
06  V<D> relaxed_convolution(int N, auto f) {
3b    V<D> R(N, 0), A(N), B(N), C;
ec    D c = 0;
af    REP(i, N) {
c0      tie(A[i], B[i]) = f(i, c);
d0      for (int p = 1; (i + 2) % p == 0; p *= 2) {
2a        if (p == i + 2) break;
bb        auto calc = [&](int la, int ra, int lb, int rb) {
58          C = multiply(V<D>(A.begin() + la, A.begin() + ra),
a1                       V<D>(B.begin() + lb, B.begin() + rb));
2b          REP(j, C.size())
5e          if (la + lb + j < N) R[la + lb + j] += C[j];
```

```
23      };
24      if (p * 2 != i + 2)
56        calc(p - 1, 2 * p - 1, i + 1 - p, i + 1);
c8      calc(i + 1 - p, i + 1, p - 1, 2 * p - 1);
e9    }
79    c = R[i];
3c  }
7f  return R;
ed }
```

taylor_shift.hpp

md5: 6d3b60

```
    // f(x+c)を返す
5d  template <class Mint>
97  V<Mint> taylor_shift(V<Mint> a, Mint c) {
40    int n = a.size();
ae    V<Mint> b(n + 1, 1), fi(n);
e2    Mint f = 1;
76    REP(i, n) {
95      a[i] *= f;
30      f *= i + 1;
98      b[i + 1] = b[i] * c;
4d    }
77    f = f.inv();
04    REP(i, n) b[n - 1 - i] *= fi[n - 1 - i] = f *= n - i;
44    reverse(b.begin(), b.end());
be    a = multiply(a, b);
8f    REP(i, n) fi[i] *= a[i + n];
dd    return fi;
6d  }
```

convex

dynamic_convex_hull.hpp

md5: 95e223

```
72  struct Line {  // f(x) = k * x + m
14    mutable ll k, m, p;
0c    bool operator<(const Line& o) const { return k < o.k; }
0d    bool operator<(ll x) const { return p < x; }
7e  };
74  struct LineContainer : multiset<Line, less<>> {
      // (for doubles, use inf = 1/.0, div(a,b) = a/b)
4a    const ll inf = LLONG_MAX;
5b    ll div(ll a, ll b) {  // floored division
3b      return a / b - ((a ^ b) < 0 && a % b);
ae    }
65    bool isect(iterator x, iterator y) {
ef      if (y == end()) {
bc        x->p = inf;
b1        return false;
1f      }
2f      if (x->k == y->k)
34        x->p = x->m > y->m ? inf : -inf;
2f      else
2c        x->p = div(y->m - x->m, x->k - y->k);
b5      return x->p >= y->p;
53    }
de    void add(ll k, ll m) {
f7      auto z = insert({k, m, 0}), y = z++, x = y;
81      while (isect(y, z)) z = erase(z);
60      if (x != begin() && isect(--x, y))
37        isect(x, y = erase(y));
a6      while ((y = x) != begin() && (--x)->p >= y->p)
ce        isect(x, erase(y));
da    }
0f    ll query(ll x) {  // max f(x)
ba      assert(!empty());
e5      auto l = *lower_bound(x);
63      return l.k * x + l.m;
2f    }
95  };
```

monge_d_edge_shortest.hpp

md5: 89b418

```
    // Int f(int a, int b){ return A[a][b] } (A : monge)
    // void det(int i, int j, Int x){
    //     argmin[i] = j; mimval[i] = x; }
2d  template <class Int, class F1, class F2>
92  void easy_larsch(int n, F1 f, F2 det, Int inf) {
3c    V<int> a(n, 0);
22    V<Int> b(n, inf);
15    n--;
eb    auto check = [&](int i, int k, int d) {
72      Int v = f(i, k);
ae      if (v < b[i]) a[i] = k, b[i] = v;
ef    };
1b    auto dfs = [&](auto& dfs, int l, int r, int d) -> void {
2e      if (r - l == 1) return (void)det(r, a[r], b[r]);
45      int m = (l + r) / 2;
fc      for (int k = a[l]; k <= a[r]; k++) check(m, k, d);
8e      dfs(dfs, l, m, d + 1);
b5      for (int k = l + 1; k <= m; k++) check(r, k, d);
bd      dfs(dfs, m, r, d + 1);
32    };
4f    check(n, 0, 0);
9c    dfs(dfs, 0, n, 1);
9a  }
    // Int f(int a, int b){ return dist(a,b) } (dist : monge)
62  template <class Int, class F1>
db  Int d_edge_shortest_path(int n, ll k, F1 f, Int max_weight,
44                           Int INF) {
da    Int inf = max_weight;  // 辺の重みより大きい値
70    auto DC = [&](Int p) -> pair<Int, ll> {
e4      vector<Int> D(n, INF);
97      vector<int> P(n);
f1      D[0] = 0;
20      easy_larsch(
20          n,
26          [&](int a, int b) { return D[b] + f(b, a) - p; },
2f          [&](int a, int b, Int c) { P[a] = b, D[a] = c; },
74          INF);
a6      ll c = 0, v = n - 1;
86      for (; v; v = P[v]) c++;
a2      return {D[n - 1], c};
13    };
ef    Int m = -inf + 1;  // 辺の重みが正なら0にしてもいい
5f    Int ok = m, ng = inf;
69    while (abs(ok - ng) > 1) {
1d      auto mid = (ok + ng) / 2;
6b      auto [d, c] = DC(mid);
86      (c <= k ? ok : ng) = mid;
b8    }
a7    if (ok == m) return -INF;
36    auto [d, c] = DC(ok);
f6    return d + ok * k;
89  }
```

cht_monotone.hpp

md5: 91223f

```
36  template <class Pos, class F>
ce  struct CHTMonotone {
45    struct Q {
2e      Pos l, r;
31      F f;
1b    };
46    deque<Q> a;
bf    CHTMonotone(Pos l, Pos r, F inf) : a({{l, r, inf}}) {}
d7    void add(F f, auto fx, auto cross) {
9b      Pos l = a[0].l, r = a.back().r;
67      Q qf = {l, r, f};
f5      if (fx(f, r) < fx(a.back().f, r)) {
c6        do {
ea          Q& q = a.back();
d7          if (fx(f, q.l) < fx(q.f, q.l)) continue;
d5          cross(q.f, f, q.r = q.l, qf.l = r);
9d          break;
e7        } while (a.pop_back(), a.size());
af        a.push_back(qf);
6e      } else if (fx(f, l) < fx(a[0].f, l)) {
93        do {
27          Q& q = a.front();
3e          if (fx(f, q.r) < fx(q.f, q.r)) continue;
26          cross(f, q.f, qf.r = l, q.l = q.r);
10          break;
bc        } while (a.pop_front(), a.size());
7b        a.push_front(qf);
e5      }
11    }
33    F query(Pos x) {
84      return ranges::partition_point(a,
e3        [x](Q& f){ return f.r < x; })->f;
56    }
50    F query_inc(Pos x) {
f0      while (a[0].r < x) a.pop_front();
3d      a[0].l = x;
```

```
6e     return a[0].f;
40   }
36   F query_dec(Pos x) {
25     while (x < a.back().l) a.pop_back();
10     a.back().r = x;
ab     return a.back().f;
e6   }
91 };
   // struct F { ll a, b; };
   // auto fx = [&](F f, ll x){ return f.a * x + f.b; };
   // void cross(F a, F b, ll& xa, ll& xb){
   //   while(abs(xa - xb) > 1){
   //     Pos x = xa + (xb - xa) / 2;
   //     (a(x) < b(x) ? xa : xb) = x;
   //   }
   // }
   // void cross(F a, F b, double& xa, double& xb){
   //   xa = xb = -(a.b - b.b) / (a.a - b.a);
   // }
```

graph

dominator_tree.hpp

md5: a10eaa

```
b7 struct DominatorTree {
60   int n, cs;
1d   V<V<int>> E, RE, rdom;
4b   V<int> S, RS, par, val, sdom, rp, idom;
fa   DominatorTree(int n)
0b       : n(n), cs(0), E(n), RE(n), rdom(n) {
5b     S = RS = par = val = sdom = rp = idom = V<int>(n, -1);
81   }
74   void add_edge(int x, int y) { E[x].push_back(y); }
79   int find(int x, int c = 0) {
88     if (par[x] == x) return c ? -1 : x;
1b     int &w = par[x], p = find(w, 1);
20     if (p < 0) return c ? w : val[x];
be     if (sdom[val[x]] > sdom[val[w]]) val[x] = val[w];
c4     w = p;
f1     return c ? p : val[x];
37   }
db   void dfs(int x) {
9f     par[cs] = sdom[cs] = val[cs] = cs;
89     RS[S[x] = cs++] = x;
9e     for (int e : E[x]) {
80       if (S[e] < 0) dfs(e), rp[S[e]] = S[x];
12       RE[S[e]].push_back(S[x]);
f6     }
b6   }
27   V<int> solve(int src) {  // idom[v] : dom tree で v の親
07     dfs(src);
45     for (int i = cs - 1; i >= 0; i--) {
2b       for (int e : RE[i]) chmin(sdom[i], sdom[find(e)]);
06       if (i) rdom[sdom[i]].push_back(i);
99       for (int e : rdom[i]) {
62         int p = find(e);
7a         idom[RS[e]] = RS[sdom[p] == i ? i : p];
e7       }
dd       if (i) par[i] = rp[i];
df     }
2a     REP(i, cs)
7b     if (i && RS[sdom[i]] != idom[RS[i]])
2c       idom[RS[i]] = idom[idom[RS[i]]];
54     return idom;
d1   }
a1 };
```

assignment_problem.hpp

md5: a097c3

```
   // result[i] : use A[i][result[i]]
e2 using Cost = ll;
8b V<int> AssignmentProblem(int n, int m, const V<V<Cost>>& A,
67                          int K, Cost INF) {
63   V<Cost> d(m + 1, 0), h(m, 0);
50   V<int> l(n, -1), r(m, -1);
a7   while (K--) {
79     V<int> f(m), par(m, -1);
4e     auto ch = [&](int w, Cost f, int v) {
47       if (f < d[w]) d[w] = f, par[w] = v;
8a     };
f7     REP(v, m + 1) d[v] = INF;
fa     REP(v, n)
b7       if (l[v] < 0) REP(w, m) ch(w, A[v][w] - h[w], v);
f3     int w = -1;
71     while (1) {
4e       w = m;
8e       REP(i, m) if (!f[i] && d[i] < d[w]) w = i;
01       f[w] = 1;
34       if (r[w] < 0) break;
5c       int v = r[w];
85       auto c = d[w] + h[w] - A[v][w];
44       REP(x, m) if (!f[x]) ch(x, c + A[v][x] - h[x], v);
f0     }
0c     REP(i, m) if (f[i]) h[i] += d[i] - d[w];
0e     while (w >= 0) swap(w, l[r[w] = par[w]]);
cb   }
fd   return l;
a0 }
```

biconnectivity.hpp

md5: 64967f

```
17 struct Biconnectivity {
2d   V<pair<int, int>> bct;  // ( 2-連結成分の番号, 頂点 )
d3   V<int> tecc;  // tecc[頂点] == 2辺連結成分の番号
a7   int num_bcc, num_tecc;
5b   Biconnectivity(int n, const V<V<int>>& g) : tecc(n) {
a9     V<int> low(n), ord(n, -1), st(n), est(n + 1, n);
92     int it = 0, idx = 0, eit = 0, f = 0;
e5     auto dfs = [&](auto& dfs, int v, int p) -> void {
96       st[it] = est[eit++] = v;
ba       low[v] = ord[v] = it++;
1e       for (int w : g[v])
3f         if (w != p || !(p = -1)) {
2f           if (ord[w] < 0) {
ed             dfs(dfs, w, v);
a3             low[v] = min(low[v], low[w]);
2a             if (low[w] >= ord[v]) {
e4               while (ord[w] < it)
cc                 bct.push_back({idx, st[--it]});
56               bct.push_back({idx++, v});
f5             }
ef           }
33           low[v] = min(low[v], ord[w]);
da         }
b4       if (low[v] == ord[v] && ++f)
46         while (est[eit] != v) tecc[est[--eit]] = f - 1;
31     };
3b     REP(v, n) if (ord[v] < 0) dfs(dfs, v, -1);
6f     REP(v, n) if (!g[v].size()) bct.push_back({idx++, v});
ae     num_bcc = idx;
79     num_tecc = f;
1c   }
64 };
```

cliques.hpp

md5: 3739d9

```
   // 計算量は O( V^2 + V 2 ^ sqrt(2E) )
   // 空集合は含まれない
   // void f(const V<int>& clique);
25 void cliques(int n, V<V<int>> graph, auto f) {
07   V<V<int>> g(n, V<int>(n));
2d   REP(x, n) for (auto y : graph[x]) g[x][y] = 1;
b9   V<int> deg(n);
a2   REP(i, n) REP(j, n) deg[i] += g[i][j];
e6   REP(it, n) {
71     int v = 0;
69     REP(i, n) if (deg[i] < deg[v]) v = i;
14     V<int> H, C = {v}, F(deg[v]);
ec     for (int w : graph[v])
ce       if (deg[w]-- < n) H.push_back(w);
66     REP(i, deg[v]) REP(j, i) F[j] |= g[H[i]][H[j]] << i;
aa     auto dfs = [&](auto& dfs, int m) -> void {
13       f(C);
05       REP(i, deg[v]) if (m & 1 << i) {
a6         C.push_back(H[i]);
bf         dfs(dfs, (m -= 1 << i) & F[i]);
f9         C.pop_back();
6f       }
79     };
21     dfs(dfs, (1 << deg[v]) - 1);
83     deg[v] = n * 5;
d7   }
37 }
```

directed_mst.hpp

md5: 4bc9ce

```
cb  template <typename T, bool isMin = true>
bd  struct LeftistHeap {
ec    struct Node {
9d      T key, lazy;
cf      Node *l, *r;
f1      int s;
98      int idx;
e3      Node(T key, int idx)
7e          : key(key), lazy(0), l(0), r(0), s(1), idx(idx) {}
61    };
db    using np = Node*;
36    np alloc(T key, int idx) { return new Node(key, idx); }
2f    np add(np t, T z) {
26      if (t) t->lazy += z, t->key += z;
c8      return t;
39    }
a9    np prop(np t) {
99      t->l = add(t->l, t->lazy);
8d      t->r = add(t->r, t->lazy);
a5      t->lazy = 0;
6d      return t;
be    }
43    np meld(np a, np b) {
32      if (!a || !b) return a ? a : b;
ca      a = prop(a), b = prop(b);
32      if ((a->key < b->key) ^ isMin) swap(a, b);
bb      a->r = meld(a->r, b);
79      if (!a->l || a->l->s < a->r->s) swap(a->l, a->r);
ea      a->s = (a->r ? a->r->s : 0) + 1;
82      return a;
4d    }
49    np pop(np t) { return meld(t->l, t->r); }
bf  };

b8  struct Edge {
40    using Cost = ll;
2b    int from, to;
27    Cost cost;
f1  };
    // root -----> leaf
    // from -----> to
    // ans[root] = -1; ans[v] = (? --> v) なる辺番号
73  V<int> directed_mst(int n, int root, V<Edge> E) {
ac    REP(i, n) if (i != root) E.push_back({(int)i, root, 0});
86    int x = 0;
3a    V<int> par(2 * n, -1), vis(par), uf(par), st;
3f    using Heap = LeftistHeap<Edge::Cost, 1>;
c0    Heap h;
70    V<typename Heap::np> ins(2 * n, 0);
07    REP(i, E.size())
23    ins[E[i].to] =
69        h.meld(ins[E[i].to], h.alloc(E[i].cost, i));
d2    auto Find = [&](auto& f, int a) -> int {
ea      return uf[a] < 0 ? a : (uf[a] = f(f, uf[a]));
24    };
e4    auto go = [&](int v) {
3b      return Find(Find, E[ins[v]->idx].from);
45    };
49    for (int i = n; ins[x]; i++) {
56      for (; vis[x] == -1; x = go(x)) vis[x] = 0;
cb      for (; x != i; x = go(x)) {
d6        auto w = ins[x]->key;
4d        ins[i] = h.meld(ins[i], h.add(h.pop(ins[x]), -w));
13        uf[x] = par[x] = i;
62      }
49      for (; ins[x] && go(x) == x; ins[x] = h.pop(ins[x]))
71        ;
4e    }
f5    V<int> ans(n, -1);
aa    for (int i = root; i != -1; i = par[i]) vis[i] = 1;
64    do {
ba      if (vis[x] != 1) {
a0        int e = ins[x]->idx;
63        ans[E[e].to] = e;
aa        for (int j = E[e].to; j != -1 && !vis[j]; j = par[j])
63          vis[j] = 1;
41      }
db    } while (x--);
65    return ans;
4b  }
```

eulerian_trail.hpp

md5: 625dd3

```
    // オイラー路を一つ構築して { 頂点列 , 辺列 } を返す
43  pair<V<int>, V<int>> eulerian_trail(
4e      int N, const V<pair<int, int>>& edges, bool directed,
9d      int start = -1) {
b1    int M = edges.size();
14    if (!M) return {{0}, {}};
e9    V<int> X(M), D(N);
b7    V<V<int>> G(N);
b8    REP(m, M) {
37      auto [u, v] = edges[m];
21      G[u].push_back(m);
c5      if (!directed) G[v].push_back(m);
19      X[m] = u + v;
69    }
53    if (start < 0) {
bb      start = edges[0].first;
01      for (auto [u, v] : edges) {
d3        D[u] += 1;
ec        D[v] += directed ? -1 : 1;
95      }
ad      REP(i, N) if (D[i] % 2 == 1) start = i;
ba    }
88    V<int> H(M + 1, start), E(M), U(M, 0), K(N);
73    REP(i, N) K[i] = G[i].size();
0e    int p = 0, q = M;
b6    while (q) {
fa      int v = H[p];
7e      while (0 < K[v]-- && U[G[v][K[v]]])
9d        ;
05      if (K[v] < 0) {
ed        if (!p) return {{-1}, {-1}};
6d        H[q--] = H[p--];
97        E[q] = E[p];
10      } else {
09        auto e = G[v][K[v]];
86        U[e] = 1;
21        E[p++] = e;
e6        H[p] = X[e] - v;
e5      }
13    }
2f    REP(i, M)
c0    if (X[E[i]] != H[i] + H[i + 1]) return {{-1}, {-1}};
d8    return {H, E};
62  }
```

manhattan_mst.hpp

md5: ea5b58

```
    // 候補の辺を O(N) 本に減らす。MST時は追加でsort,
    // UF等の必要あり。
67  template <typename T>
9a  V<tuple<T, int, int>> manhattan_mst(V<T> xs, V<T> ys) {
52    assert(xs.size() == ys.size());
c1    V<tuple<T, int, int>> ret;
5d    int n = (int)xs.size();
f6    V<int> ord(n);
be    REP(i, n) ord[i] = i;
65    auto cmp = [&](int i, int j) {
b9      return xs[i] + ys[i] < xs[j] + ys[j];
6d    };
4d    REP(s, 4) {
d3      sort(ord.begin(), ord.end(), cmp);
10      map<T, int> idx;
8f      for (int i : ord) {
d1        for (auto it = idx.lower_bound(-ys[i]);
b6             it != idx.end(); it = idx.erase(it)) {
06          int j = it->second;
3e          if (xs[i] - xs[j] < ys[i] - ys[j]) break;
63          ret.emplace_back(
0c              abs(xs[i] - xs[j]) + abs(ys[i] - ys[j]), i, j);
50        }
b2        idx[-ys[i]] = i;
60      }
d6      swap(xs, ys);
b7      if (s % 2) REP(i, n) xs[i] *= -1;
b7    }
99    return ret;
ea  }
```

max_flow.hpp

md5: 0e6583

```
   /*
   struct E {
     int to, rev, cap;
   };
   V<V<E>> g;
   auto add_edge = [&](int from, int to, int cap) {
     g[from].push_back(E{to, int(g[to].size()), cap});
     g[to].push_back(E{from, int(g[from].size())-1, 0});
   };
   */
40 template <class C>
ab struct MaxFlow {
2a   C flow;
d3   V<char> dual;  // false: S-side true: T-side
b3 };
6d template <class C, class E>
8e struct MFExec {
cd   static constexpr C INF = numeric_limits<C>::max();
91   C eps;
a8   V<V<E>>& g;
e0   int s, t;
c5   V<int> level, iter;
26   C dfs(int v, C f) {
ae     if (v == t) return f;
15     C res = 0;
64     for (int& i = iter[v]; i < int(g[v].size()); i++) {
d7       E& e = g[v][i];
60       if (e.cap <= eps || level[v] >= level[e.to])
14         continue;
3d       C d = dfs(e.to, min(f, e.cap));
b2       e.cap -= d;
36       g[e.to][e.rev].cap += d;
63       res += d;
26       f -= d;
04       if (f == 0) break;
24     }
ff     return res;
50   }
bb   MaxFlow<C> info;
26   MFExec(V<V<E>>& _g, int _s, int _t, C _eps)
59       : eps(_eps), g(_g), s(_s), t(_t) {
ee     int N = int(g.size());
a1     C& flow = (info.flow = 0);
b6     while (true) {
d5       queue<int> que;
37       level = V<int>(N, -1);
db       level[s] = 0;
75       que.push(s);
39       while (!que.empty()) {
fc         int v = que.front();
bc         que.pop();
c5         for (E e : g[v]) {
91           if (e.cap <= eps || level[e.to] >= 0) continue;
d1           level[e.to] = level[v] + 1;
3d           que.push(e.to);
d4         }
2c       }
f2       if (level[t] == -1) break;
2d       while (true) {
d2         iter = V<int>(N, 0);
86         C f = dfs(s, INF);
6c         if (!f) break;
0c         flow += f;
5f       }
4d     }
e0     for (int i = 0; i < N; i++)
1e       info.dual.push_back(level[i] == -1);
4f   }
f4 };
4e template <class C, class E>
7a MaxFlow<C> get_mf(V<V<E>>& g, int s, int t, C eps) {
7b   return MFExec<C, E>(g, s, t, eps).info;
0e }
```

max_matching-daimonge.hpp

md5: aa4a41

```
aa V<pair<ll, ll>> MaxMatching(ll n, V<pair<ll, ll>> &es) {
50   mt19937 rnd;
f8   V<V<ll>> g(n + 1);
bb   for (auto &[u, v] : es) {
52     g[u + 1].push_back(v + 1);
95     g[v + 1].push_back(u + 1);
c5   }
cd   V<ll> used(n + 1), mate(n + 1);
ce   ll T;
29   auto dfs = [&](auto &dfs, ll v) -> bool {
d7     used[v] = T;
ca     shuffle(g[v].begin(), g[v].end(), rnd);
15     for (auto &u : g[v]) {
8e       ll w = mate[u];
69       if (used[w] < T) {
ed         mate[v] = u, mate[u] = v, mate[w] = 0;
06         if (!w or dfs(dfs, w)) return 1;
c4         mate[u] = w, mate[w] = u, mate[v] = 0;
26       }
58     }
a5     return 0;
50   };
4b   REP(_, 10) {
7e     used.assign(n + 1, 0);
b8     REP(v, n) if (!mate[v + 1]) T++, dfs(dfs, v + 1);
49   }
ca   V<pair<ll, ll>> ret;
84   REP(v, n)
94   if (v + 1 < mate[v + 1])
62     ret.push_back({v, mate[v + 1] - 1});
11   return ret;
aa }
```

min_cost_flow.hpp

md5: 3550f7

```
   // 辺は配列 E に一列に並んでいる。
   // 辺 e (e : int) の逆辺は、辺 e^1 である。
dc struct MCF {
1c   using Cap = ll;
3a   using Cost = ll;
a0   Cost INF = 1ll << 60;
c5   struct Edge {
cf     int from;
e2     int to;
86     Cost cost;
06     Cap cap;
fc   };
22   int N;
77   V<Edge> E;
37   V<V<int>> g;
c6   V<Cost> P;
6e   MCF(int n) : N(n), g(n), P(n, 0) {}
b9   int addEdge(int u, int v, Cap cap, Cost cost) {
01     int e = E.size();
42     g[u].push_back(e);
7b     g[v].push_back(e + 1);
c5     E.push_back({u, v, cost, cap});
d2     E.push_back({v, u, -cost, 0});
e9     return e;
fc   }
01   pair<Cap, Cost> flowuni(int s, int t, Cap lim) {
a7     V<Cost> dist(N, INF);
8a     V<int> par(N, -1);
45     V<Cap> fl(N, lim);
85     priority_queue<pair<Cost, int>> que;
e9     dist[s] = 0;
08     que.push({0, s});
bd     while (que.size()) {
8c       auto [d, v] = que.top();
6a       que.pop();
98       if (dist[v] != -d) continue;
19       if (v == t) break;
bd       for (auto e : g[v])
1b         if (E[e].cap) {
97           int w = E[e].to;
ac           Cost nd = -d + P[v] + E[e].cost - P[w];
df           if (dist[w] <= nd) continue;
83           dist[w] = nd;
e3           que.push({-nd, w});
28           par[w] = e;
ee           fl[w] = min(fl[v], E[e].cap);
8d         }
37     }
4b     if (par[t] < 0) return {0, 0};
11     lim = fl[t];
82     for (int p = t; p != s;) {
ca       int e = par[p];
```

```
7f        E[e].cap -= lim;
a7        E[e ^ 1].cap += lim;
dd        p = E[e ^ 1].to;
16      }
4e      for (int i = 0; i < N; i++)
14        if (i == s || par[i] >= 0) {
f9          P[i] += dist[i] - dist[t];
6e        }
00      return {P[t] - P[s], lim};
d7    }
f8    V<pair<Cap, Cost>> slope(int s, int t, Cap lim) {
92      V<pair<Cap, Cost>> res;
d4      res.push_back({0, 0});
04      Cap fl = 0;
2f      Cost co = 0;
f7      while (fl < lim) {
87        auto [u, v] = flowuni(s, t, lim - fl);
f0        if (v == 0) break;
df        res.push_back({fl += v, co += u * v});
69      }
0a      return res;
95    }
fd    pair<Cap, Cost> flow(int s, int t, Cap lim) {
b8      return slope(s, t, lim).back();
bb    }
35  };
```

scc.hpp

md5: 467325

```
    // res.first = 強連結成分の個数
    // res.second[v] = v が属する強連結成分の番号
be  pair<int, V<int>> scc(int n, V<V<int>> g) {
e1    int tm = 0, it = 0, idx = 0;
e8    V<int> ord(n, -1), st(n + 1, n), id(n, n);
16    auto dfs = [&](auto& dfs, int v) -> int {
98      int low = ord[v] = tm++;
ba      st[it++] = v;
39      for (auto w : g[v])
19        if (id[w] >= n)
6e          chmin(low, ord[w] < 0 ? dfs(dfs, w) : ord[w]);
7a      if (low == ord[v] && ++idx)
f4        while (st[it] != v) id[st[--it]] = idx;
1e      return low;
c3    };
06    REP(i, n) if (ord[i] == -1) dfs(dfs, i);
98    for (auto& x : id) x = idx - x;
09    return make_pair(idx, move(id));
46  }
```

two-sat.hpp

md5: 9301c7

```
d9  struct TwoSat {
e5    int n;
3a    V<bool> res;
f3    V<V<int>> g;
81    TwoSat(int n) : n(n), g(n * 2) {}
cb    void add(int x) { g.resize((n += x) * 2); }
47    void either(ll i, bool fi, ll j, bool fj) {
87      i = i * 2 + fi, j = j * 2 + fj;
96      g[i ^ 1].push_back(j);
40      g[j ^ 1].push_back(i);
f6    }
fe    bool solve() {
e7      auto id = scc(g.size(), g).second;
9e      res.resize(n);
16      REP(i, n) if (id[i * 2] == id[i * 2 + 1]) return 0;
35      REP(i, n) res[i] = id[i * 2] < id[i * 2 + 1];
b2      return 1;
f6    }
09    void atMostOne(V<pair<ll, bool>> a) {
b4      ll x = n, i = 0;
18      add(a.size());
1c      for (auto [v, f] : a) {
af        either(v, !f, x + i, 1);
42        if (i) either(x + i - 1, 0, x + i, 1);
14        if (i) either(v, !f, x + i - 1, 0);
58        i++;
11      }
8b    }
93  };
```

tree

hld.hpp

md5: cd5ee4

```
40  struct HLD {
5b    int N;
44    int g;
e6    V<V<int>> t;
88    V<int> par, dep, sz, pp, stov, vtos, px, index;
4c    HLD(int N, V<V<int>> graph, int root)
7b        : N(N), g(0), t(move(graph)), par(N, -1), dep(N),
88          sz(N, 1), pp(N), stov(N), vtos(N), px(N),
44          index(N) {
56      dfs1(root);
3e      dfs2(root);
12    }
f6    void dfs1(int v) {
81      for (auto& w : t[v]) {
b9        t[w].erase(find(t[w].begin(), t[w].end(), v));
7b        par[w] = v;
85        dep[w] = dep[v] + 1;
90        dfs1(w);
4d        sz[v] += sz[w];
1e        if (sz[t[v][0]] < sz[w]) swap(t[v][0], w);
86      }
fc    }
40    void dfs2(int v) {
ef      vtos[v] = g;
98      stov[g++] = v;
b9      for (auto w : t[v]) {
c4        int f = w == t[v][0];
85        pp[w] = f ? pp[v] : w;
d7        px[w] = px[v] + 1 - f;
f7        dfs2(w);
65      }
89    }
53    int lca(int a, int b) {
0d      if (px[pp[a]] > px[pp[b]]) swap(a, b);
d5      while (px[pp[a]] < px[pp[b]]) b = par[pp[b]];
fe      while (pp[a] != pp[b]) {
cc        a = par[pp[a]];
8a        b = par[pp[b]];
b5      }
ad      return dep[a] < dep[b] ? a : b;
82    }
11    auto query_half(int r, int a, bool edge = false) {
bd      vector<pair<int, int>> res(px[a] - px[r] + 1);
35      while (px[a] > px[r]) {
a6        res[px[a] - px[r]] = {vtos[pp[a]], vtos[a] + 1};
45        a = par[pp[a]];
1c      }
fe      res[0] = {vtos[r] + edge, vtos[a] + 1};
58      return res;
a2    }
      // return vector<pair<int,int>>
      // a -> b
      // l<rなら[l:r)、l>rなら[r:l)の逆順
      // edge=trueの場合はlcaを含まない区間を返す
8c    auto query(int a, int b, bool edge = false) {
b6      int g = lca(a, b);
e6      auto l = query_half(g, b, edge);
67      auto r = query_half(g, a, 1);
b8      for (auto& q : r) swap(q.first, q.second);
05      l.insert(l.begin(), r.rbegin(), r.rend());
3f      return l;
92    }
      // 返り値は縮約後の縮約後の頂点集合 vector<int> と
      // 木の辺の集合 vector<pair<int,int>> !! もとの頂点番号
      // !! もし reindex したいなら index[v] を見ればいい
e6    pair<V<int>, V<pair<int, int>>> ComputeTree(V<int> vs) {
0e      auto comp = [&](int x, int y) {
16        return vtos[x] < vtos[y];
08      };
71      sort(vs.begin(), vs.end(), comp);
ad      int K = vs.size();
a0      REP(i, K - 1) vs.push_back(lca(vs[i], vs[i + 1]));
e4      sort(vs.begin(), vs.end(), comp);
22      vs.erase(unique(vs.begin(), vs.end()), vs.end());
0c      K = vs.size();
0f      REP(i, K) index[vs[i]] = i;
cc      V<pair<int, int>> es;
1b      REP(i, K - 1)
```

```
c2      es.push_back({lca(vs[i], vs[i + 1]), vs[i + 1]});
c5      return {vs, es};
4d    }
cd  };
```

link_cut_tree.hpp

md5: 20f709

```
75  struct S {
63    ll x;
41  };
94  static S op(S l, S r) { return {l.x + r.x}; }
10  struct LCNode {
43    LCNode *c[2] = {0, 0}, *p = 0;
f4    S a, sum;
78    int rev = 0;
09    void do_rev() { rev ^= 1, swap(c[0], c[1]), agg(); }
98    void prop() {
a7      if (rev) REP(t, 2) if (c[t]) c[t]->do_rev();
85      rev = 0;
32    }
65    void agg() {
90      sum = a;
96      if (c[0]) sum = op(c[0]->sum, sum);
2f      if (c[1]) sum = op(sum, c[1]->sum);
69    }
42    int way() {
f1      REP(t, 2) if (p && p->c[t] == this) return t;
59      return -1;
0f    }
81    void rot() {
7e      auto w = p;
4a      int x = way();
ea      if (w->way() != -1) w->p->c[w->way()] = this;
ab      p = w->p;
d6      w->c[x] = c[!x];
a8      if (c[x ^ 1]) c[!x]->p = w;
49      c[!x] = w;
a9      w->p = this;
23      w->agg(), agg();
16    }
f2    void propall() {
37      if (way() >= 0) p->propall();
22      prop();
ea    }
16    void splay() {
ab      for (propall(); way() >= 0; rot())
ef        if (p->way() != -1)
4b          (way() == p->way() ? p : this)->rot();
0c    }
58    void expose() {
f1      for (auto t = this; t; t = t->p) t->splay();
46      for (auto t = this; t->p; t = t->p) {
ec        t->p->c[1] = t;
50        t->p->agg();
86      }
78      splay(), c[1] = 0, agg();
6d    }
94    void evert() { expose(), do_rev(); }
3a    void link(LCNode* r) { r->evert(), evert(), p = r; }
e1    void cut() { expose(), c[0] = c[0]->p = 0; }
20  };
```

static_top_tree.hpp

md5: 4a3228

```
3a  struct StaticTopTree {
e5    static const int VERT = 0;  // -1      -1      vtx
d3    static const int COMP = 1;  // root    child   eidx+n
10    static const int RAKE = 2;  // child   child   -1
3c    static const int ADDE = 3;  // child   -1      eidx+n
1f    static const int ADDV = 4;  // child   -1      vtx
9b    using pr = pair<int, int>;
4e    int n, rt, k, num_node;
37    V<V<pr>> g;
3f    V<int> PE, vh;
fa    struct Node {
18      int p, l, r, e, t;
f5    };
55    V<Node> A;
fd    StaticTopTree() {}
d3    StaticTopTree(int n, const V<pr>& tree, int root)
e3        : n(n), rt(root), k(0), g(n), PE(n, -1),
f8          vh(n * 2, -1), A(4 * n + 30, {-1, -1, -1, 0, -1}) {
a4      REP(i, n - 1) {
0b        auto [u, v] = tree[i];
fe        g[u].push_back({v, i});
4d        g[v].push_back({u, i});
3e      }
3b      dfs(rt);
81      A.resize(num_node = comp(rt).first + 1);
25    }
9c    int dfs(int c) {
2c      int s = 1, best = 0, t = 0;
a4      for (auto& e : g[c]) {
a5        auto [w, ei] = e;
03        g[w].erase(ranges::find(g[w], pr{c, PE[w] = ei}));
b2        s += t = dfs(w);
0e        if (best < t) best = t, swap(e, g[c][0]);
2a      }
13      return s;
a0    }
99    int add(int l, int r, int e, int t) {
45      A[k] = {-1, l, r, e, t};
4c      if (l >= 0) A[l].p = k;
1e      if (r >= 0) A[r].p = k;
95      if (e >= 0) vh[e] = k;
cb      return k++;
cb    }
da    pr merge(const V<pr>& a, int t) {
91      if (a.size() == 1) return a[0];
c8      int u = 0;
2c      for (auto& [_, s] : a) u += s;
22      V<pr> b, c;
68      for (auto& [i, s] : a) {
b2        (u > s ? b : c).emplace_back(i, s);
5e        u -= s * 2;
74      }
9e      int e = t == 1 ? n + PE[A[c[0].first].e] : -1;
b8      return {
96          add(merge(b, t).first, merge(c, t).first, e, t),
0c          -u};
95    }
30    pr comp(int v) {
b4      V<pr> chs{addv(v)};
e9      while (g[v].size())
6a        chs.push_back(addv(v = g[v][0].first));
80      return merge(chs, 1);
cb    }
a2    pr rake(int v) {
69      if (g[v].size() < 2) return {-1, 0};
1b      V<pr> chs;
6a      REP(i, g[v].size() - 1)
fa      chs.push_back(adde(g[v][i + 1].first));
ab      return merge(chs, 2);
d7    }
a4    pr adde(int v) {
9a      auto [j, sj] = comp(v);
63      return {add(j, -1, n + PE[v], 3), sj};
7e    }
93    pr addv(int v) {
38      auto [i, sj] = rake(v);
1c      return {add(i, -1, v, i < 0 ? 0 : 4), ++sj};
a0    }
33    auto& operator[](int i) const { return A[i]; }
92  };
62  template <class Structure>
3e  struct STTAgg {
cf    Structure d;
23    V<class Structure::Node> A;
50    StaticTopTree st;
76    void agg(int v) {
ed      auto [p, l, r, e, t] = st[v];
40      if (t == 0) A[v].s = d.vert(e);
4f      if (t == 1) A[v].s = d.comp(A[l].s, e - st.n, A[r].s);
e4      if (t == 2) A[v].p = d.rake(A[l].p, A[r].p);
f1      if (t == 3) A[v].p = d.adde(e - st.n, A[l].s);
3d      if (t == 4) A[v].s = d.addv(e, A[l].p);
ab      if (t == 5) A[v].s = d.rev(A[l].s);
9a    }
68    void aggX(int v) {
a8      while (v >= 0) agg(v), v = st[v].p;
6f    }
f3    void aggV(int v) { aggX(st.vh[v]); }
3d    void aggE(int e) { aggX(st.vh[st.n + e]); }
4b    void init() {
6e      A.resize(st.num_node);
a1      REP(i, A.size()) agg(i);
```

```
23    }
72  };
8f  struct STTStructure {
9f    struct Seg {};
d2    struct Pt {};
15    struct Node {
82      Seg s;
45      Pt p;
2e    };
a3    Seg vert(int v) { return {}; }
87    Seg comp(Seg l, int e, Seg r) { return {}; }
72    Pt rake(Pt l, Pt r) { return {}; }
6a    Seg addv(int v, Pt p) { return {}; }
33    Pt adde(int e, Seg s) { return {}; }
ed    Seg rev(Seg s) { return s; }
4a  };
    // V<pair<int,int>> edges(N-1);
    // STTAgg<STTStructureInst> ds;
    // ds.d.vtx = move(A);
    // ds.d.edge = move(E);
    // ds.st = StaticTopTree(N, edges, 0);
    // ds.init();
```

tree_dp.hpp

md5: a39ad5

```
    // F : S rake(S l, S r);
    // G : S extend(S a, int old_root, int new_root)
86  template <class S, class F, class G>
a7  struct TreeDP {
ed    int n, root;
52    V<int> par;
33    V<S> low, high;
aa    F f;
60    G g;
8c    TreeDP(V<V<int>> t, V<S> prim, F f, G g, int r, S e)
b1        : n(t.size()), root(r), par(n, -1), low(prim),
1b          high(prim), f(move(f)), g(move(g)) {
f9      V<int> bfs(n, r);
dc      int p = 1;
7c      for (int v : bfs)
07        for (int w : t[v]) {
5f          ranges::rotate(t[w], ranges::find(t[w], v) + 1);
4e          t[w].pop_back();
52          par[bfs[p++] = w] = v;
e0        }
45      REP(i, n - 1) {
21        int v = bfs[n - 1 - i], w = par[v];
46        low[w] = f(g(low[v], v, w), low[w]);
d8      }
9e      for (int v : bfs) {
07        S s = prim[v];
51        for (int w : t[v]) {
63          high[w] = s;
25          s = f(s, g(low[w], w, v));
50        }
a7        ranges::reverse(t[v]);
6b        s = v == r ? e : g(high[v], par[v], v);
7b        for (int w : t[v]) {
3a          high[w] = f(s, high[w]);
6a          s = f(g(low[w], w, v), s);
a3        }
c4      }
df    }
c4    S get(int v) {
c0      if (v == root) return low[v];
28      return f(low[v], g(high[v], par[v], v));
af    }
a3  };
```

string

z_algorithm.hpp

md5: 3956ea

```
19  template <class Iter>
63  V<int> z_algorithm(Iter s, Iter s_end) {
75    int n = s_end - s;
0d    vector<int> r(n + 1);
8f    for (int i = 1, j = 0; i <= n; i++) {
33      int& k = r[i];
87      if (i < j + r[j]) k = min(j + r[j] - i, r[i - j]);
da      while (i + k < n && s[k] == s[i + k]) k++;
1b      if (j + r[j] < i + r[i]) j = i;
05    }
18    r[0] = n;
05    return r;
39  }
```

aho-corasick-direct-link.hpp

md5: 864ee9

```
d4  struct ACNode {
61    ll len = 0, li = 0;  // suffix link
eb    array<ll, 26> nx = {};  // 1 文字足したときの最長接尾辞
57  };
61  template <class Str, ll offset = 'a'>
c0  pair<V<ACNode>, V<ll>> AhoCorasick(const V<Str>& s) {
10    V<ACNode> a(1);
79    V<ll> p(s.size()), bfs(s.size());
fb    REP(i, s.size()) bfs[i] = i;
69    REP(i, bfs.size()) {
04      ll v = bfs[i];
3a      if (a[p[v]].len == s[v].size()) continue;
43      ll &q = p[v], c = s[v][a[q].len] - offset;
cc      if (a[a[q].nx[c]].len <= a[q].len) {
21        a.push_back(ACNode());
a6        a[q].nx[c] = a.size() - 1;
65        a.back().len = a[q].len + 1;
51      }
7a      q = a[q].nx[c];
ad      bfs.push_back(v);
b0    }
cc    for (auto& q : a) REP(j, q.nx.size()) if (q.len) {
2c        if (q.nx[j]) a[q.nx[j]].li = a[q.li].nx[j];
c2        if (!q.nx[j]) q.nx[j] = a[q.li].nx[j];
1f      }
5e    return {a, p};
86  };
```

aho-corasick.hpp

md5: 308aec

```
d1  struct ACTrie {
8e    using NP = ACTrie*;
b2    V<int> acc;
e0    map<int, NP> next;
4c    NP fail = 0, dnx = 0;

2b   private:
6b    void add(const string& s, int id, int p = 0) {
d1      if (p == int(s.size())) {
b5        acc.push_back(id);
01        return;
ff      }
3b      if (!next[s[p]]) next[s[p]] = new ACTrie();
8d      next[s[p]]->add(s, id, p + 1);
a0    }
9b    template <class OP>
2a    NP count(OP op, int p) {
4e      if (!fail) return this;
23      for (int id : acc) op(id, p);
53      if (dnx) {
50        dnx->count(op, p);
4a      } else {
54        dnx = fail->count(op, p);
bd      }
2f      return acc.size() ? this : dnx;
36    }

c7   public:
      // パターンにマッチするたびにop(string ID,
      // 発見位置の終端)を呼び出す
      // 終端が同じで複数マッチする文字列が存在する場合, 長い順に呼び出さ
れる
      // s = "abaaba", pattern = {"ab", "ba"} なら
      // op(0, 2), op(1, 3), op(0, 5), op(1, 6)
8c    template <class OP>
e4    void match(const string& s, OP op, int p = 0) {
a5      if (p == int(s.size())) return;
05      if (next[s[p]]) {
c7        next[s[p]]->count(op, p + 1);
d3        next[s[p]]->match(s, op, p + 1);
c3      } else {
f2        if (!fail)
59          match(s, op, p + 1);  // root
cb        else
ad          fail->match(s, op, p);  // other
96      }
```

```
54    }
b5    static NP make(V<string> v) {
90      NP tr = new ACTrie();
cb      for (int i = 0; i < int(v.size()); i++) {
44        tr->add(v[i], i);
12      }
2c      V<NP> q = {tr};
c7      tr->fail = 0;
82      for (size_t qi = 0; qi < q.size(); qi++) {
56        NP ntr = q[qi];
48        for (auto [i, to] : ntr->next) {
67          NP f = ntr->fail;
02          while (f && !f->next.count(i)) f = f->fail;
1d          to->fail = f ? f->next[i] : tr;
cb          q.push_back(to);
bc        }
c3      }
ed      return tr;
11    }
30  };
```

lcp_array.hpp

md5: c04e48

```
e8  V<ll> lcp_array(auto s, V<ll> sa) {
be    ll n = s.size(), h = 0;
91    V<ll> rsa(n + 1), lcp(n);
d7    REP(i, n) rsa[sa[i]] = i;
4b    REP(i, n) {
cb      if (h > 0) h--;
a5      if (rsa[i] == n - 1) continue;
7e      ll j = sa[rsa[i] + 1];
d2      for (; j + h < n && i + h < n; h++) {
e4        if (s[j + h] != s[i + h]) break;
2b      }
e5      lcp[rsa[i]] = h;
d2    }
ba    return lcp;
c0  }
```

manacher.hpp

md5: d88928

```
    // res[i] = ( l + r == i なる回文 s[l..r) の長さの最大値 )
    // dummy は s 内で使用されていない値
f3  template <class S>
c9  V<int> manacher(const S& s, int dummy) {
a9    int n = s.size(), m = n * 2 + 1;
98    S a(m, dummy);
9a    REP(i, n) a[i * 2 + 1] = s[i];
a5    V<int> r(m);
93    r[0] = 1;
0c    for (int i = 1, j = 0; i < m; i++) {
27      int& k = r[i];
3e      k = min(j + r[j] - i, r[2 * j - i]) * (i < j + r[j]);
43      while (0 <= i - k && i + k < m && a[i - k] == a[i + k])
26        k++;
33      if (j + r[j] < i + r[i]) j = i;
51    }
e0    REP(i, m) r[i] -= 1;
e0    return r;
d8  }
```

mp.hpp

md5: ed660d

```
    // S[:i]の接頭辞と接尾辞が最大何文字一致しているか
    // S[:i]=T*cとかけるTの長さの最小値はi%mp[i]=0?mp[i]:i
f3  template <class S>
6e  V<int> mp(const S& s) {
23    int n = int(s.size());
d3    V<int> r(n + 1);
6f    r[0] = -1;
73    for (int i = 0, j = -1; i < n; i++) {
74      while (j >= 0 && s[i] != s[j]) j = r[j];
3c      j++;
75      r[i + 1] = j;
ba    }
42    return r;
ed  }
```

palindromic_tree.hpp

md5: c35e22

```
24  struct PalNode {
38    ll len = 0, diff = 0;
d8    PalNode *li = 0;   // suffix link
e7    PalNode *par = 0;  // abcba --> bcb
fe    PalNode *sli = 0;  // series link
61    PalNode *nx[26] = {};
1d  };
1a  V<PalNode> pool(1000000);  // string length + O(1)
43  PalNode *poolptr = pool.data();
02  struct PalTree {
ea    using np = PalNode *;
ca    np n0, n1, v;
39    V<int> s;
38    PalTree() : n0(poolptr++), n1(poolptr++), v(n1) {
75      *n0 = {-1};
af      *n1 = {0, 1, n0};
b8    }
3f    np add_char(int c) {
b8      int i = s.size();
82      s.push_back(c);
bd      if (v->len == i) v = v->li;
34      while (s[i - v->len - 1] != c) v = v->li;
87      if (!v->nx[c]) {
a1        np n = poolptr++, m = v->li, l = n1;
14        if (v->len >= 0) {
b6          while (s[i - m->len - 1] != s[i]) m = m->li;
a6          l = m->nx[c];
4b        }
6f        v->nx[c] = n;
9c        *n = {v->len + 2, v->len + 2 - l->len, l, v};
96        n->sli = n->diff == l->diff ? l->sli : n;
da      }
95      return v = v->nx[c];
56    }
c3  };
```

rolling-hash.hpp

md5: 0d786b

```
c4  struct RollingHash {
c4    using ull = unsigned long long;
c7    constexpr static ull MOD = (1ULL << 61) - 1;
      // a < MOD, b < MOD必須
f5    constexpr static ull add(ull a, ull b) {
41      if ((a += b) >= MOD) a -= MOD;
32      return a;
ce    }
      // a < MOD, b < MOD必須
51    constexpr static ull mul(ull a, ull b) {
40      __uint128_t c = (__uint128_t)(a)*b;
7b      return add((ull)(c >> 61), (ull)(c & MOD));
21    }
83    void expand(int n) {
bd      while (int(power.size()) <= n)
2c        power.push_back(mul(power.back(), base));
05    }

b7    unsigned base;
08    vector<ull> power;
aa    RollingHash() {
69      base = random_device{}();
94      power = {1};
cf    }
22    vector<ull> build(const auto& s) {
bf      vector<ull> res;
d1      res.reserve(ssize(s) + 1);
7f      res.push_back(0);
f4      for (auto c : s) {
bc        res.push_back(add(mul(res.back(), base), c));
0e      }
03      return res;
1d    }
61    ull query(const vector<ull>& hash, int l, int r) {
2b      expand(r - l);
8f      return add(hash[r], MOD - mul(hash[l], power[r - l]));
ca    }
0d  };
```

run_enum.hpp

md5: cea301

```
    // itereator バージョンの z_algorithm が必要
66  struct Run {
e5    int p, l, r;
19  };  // period, l, r
76  V<Run> EnumerateRuns(string S) {
80    int n = S.size();
```

```
16    V<Run> res;
91    auto f = [&](auto& f, int l, int r) -> void {
83      int len = r - l, lz = len / 2, rz = len - lz;
6c      if (len < 2) return;
d9      V<int> tg(len * 2);
50      REP(t, 2) REP(i, len) tg[t * len + i] = S[l + i];
1b      f(f, l + lz, r);
fb      f(f, l, l + lz);
10      auto a = z_algorithm(tg.begin() + lz, tg.end());
96      auto b = z_algorithm(tg.rbegin() + rz, tg.rend());
c1      auto ch = [&](int p, int lx, int rx) {
66        if (l != 0 && lx == lz) return;
6b        if (r != n && rx == rz) return;
1a        if (lx + rx < p * 2) return;
d7        res.push_back({p, l + lz - lx, l + lz + rx});
91      };
65      for (int p = 1; p <= lz; p++)
09        if (a[len - p] < p)
73          ch(p, min(lz, b[p] + p), min(rz, a[len - p]));
73      for (int p = 1; p <= rz; p++)
57        ch(p, min(lz, b[len - p]), min(rz, a[p] + p));
b5    };
be    f(f, 0, n);
07    return res;
ce  }
```

suffix_array.hpp

md5: 43ef37

```
62  V<ll> sa_is(auto s, ll B = 200) {
2d    ll n = s.size(), m = 0, b = 0;
4f    if (n <= 1) return {0};
c4    V<ll> a(n + 1, -1), sa(n), ls(n);
3b    REP(i, n) a[i] = s[i];
6d    for (ll i = n - 2; i >= 0; i--)
c8      ls[i] =
36          a[i] < a[i + 1] || a[i] == a[i + 1] && ls[i + 1];
76    auto induce = [&](const V<ll>& lms) {
ec      REP(i, n) sa[i] = -1;
4d      V<ll> zs(B + 1), zl(B + 2), yl(B + 2);
be      REP(i, n) ls[i] ? zl[a[i] + 1]++ : zs[a[i]]++;
fb      REP(i, B + 1) yl[i] = zl[i + 1] += zs[i] += zl[i];
fd      REP(i, m) sa[zs[a[lms[i]]]++] = lms[i];
2c      sa[zl[a[n - 1]]++] = n - 1;
61      for (ll v : sa)
5c        if (v-- > 0 && !ls[v]) sa[zl[a[v]]++] = v;
da      REP(i, n)
c8      if (ll v = sa[n - 1 - i]; v-- > 0 && ls[v])
4a          sa[--yl[a[v]]] = v;
de    };
8a    V<ll> f(n + 1, -1), lms(n + 1, n);
de    REP(i, n)
75    if (i && !ls[i - 1] && ls[i]) lms[f[i] = m++] = i;
35    induce(lms);
82    if (m) {
32      V<ll> res, nx(m);
b7      REP(i, n) if (f[sa[i]] >= 0) res.push_back(sa[i]);
ef      REP(i, m - 1) {
1c        ll l = res[i], r = res[i + 1];
ed        ll el = lms[f[l] + 1], er = lms[f[r] + 1];
ec        while (l < el && r < er && a[l] == a[r]) l++, r++;
5b        nx[f[res[i + 1]]] = b += l < el || r < er;
57      }
93      auto q = sa_is(nx, b);
c8      for (ll& x : q) x = lms[x];
9d      induce(q);
90    }
b2    return sa;
43  }
```

suffix_array_doubling.hpp

md5: 073492

```
24  template <class Str>
4d  V<ll> suffix_array(Str s) {
cf    ll n = s.size();
0d    V<ll> sa(n), rsa(n), tmp(n);
c1    REP(i, n) sa[i] = i;
20    REP(i, n) rsa[i] = s[i];  // 必要ならここで座圧
93    for (ll k = 1; k < n; k *= 2) {
0f      auto cmp = [&](ll x, ll y) {
8f        if (rsa[x] != rsa[y]) return rsa[x] < rsa[y];
c3        ll rx = x + k < n ? rsa[x + k] : -1;
f6        ll ry = y + k < n ? rsa[y + k] : -1;
99        return rx < ry;
31      };
3d      sort(sa.begin(), sa.end(), cmp);
2f      tmp[sa[0]] = 0;
e9      REP(i, n - 1)
bf      tmp[sa[i + 1]] = tmp[sa[i]] + cmp(sa[i], sa[i + 1]);
29      REP(i, n) rsa[i] = tmp[i];
63    }
bf    return sa;
07  }
```

suffix_automaton.hpp

md5: 0acc15

```
92  struct SuffixAutomaton {
d1    using Char = char;
cb    using Str = string;

      /*
        Node information

        構築時に作る情報を増やしたいなら、
          - Node の定義
          - MakeNode() での初期化
          - clone のときの deep copy
        を変更する

        len :
          マッチするsubstrのうちlongest (4)
      */
ad    struct Node {
4c      using NP = Node*;
19      int id;
66      int len;
62      NP link;
c9      map<Char, NP> next;

ec      Node() {}
c7      int getLongest() { return len; }
49      int getShortest() { return link ? link->len + 1 : 0; }
3e      void putNext(Char c, NP to) { next[c] = to; }
c1      bool hasNext(Char c) { return next.count(c); }
a9      NP getNext(Char c) { return hasNext(c) ? next[c] : 0; }
bc    };
c0    using NP = Node*;
15    NP last;
c0    int sz;  // the number of nodes
d8    V<NP> nodes;
0e    SuffixAutomaton(int N) : sz(0), nodes(2 * N, 0) {
17      last = nodes[0] = MakeNode(0);
69    }

4c    NP MakeNode(int len) {
98      NP v = nodes[sz] = new Node();
7d      v->id = sz++;
db      v->len = len;
b2      v->link = 0;
f8      return v;
3d    }
      /*
        Add c to nodes[last]
        return the new node id
      */
e5    void add_char(Char c) {
77      NP p = last;
ad      last = MakeNode(last->len + 1);
ca      for (; p && !p->hasNext(c); p = p->link)
97        p->next[c] = last;
19      if (!p) {
06        last->link = nodes[0];
7a      } else {
62        NP q = p->getNext(c);
60        if (p->len + 1 == q->len) {
cd          last->link = q;
64          return;
87        }
          // clone!
c9        NP w = MakeNode(p->len + 1);
          /*
            deep copy !
          */
8c        w->next = q->next;
de        w->link = q->link;
e3        for (; p && p->getNext(c) == q; p = p->link)
16          p->next[c] = w;
```

```
f9        q->link = last->link = w;
5f      }
75    }
43    void TopologicalSort() {
9a      V<int> deg(sz);
63      V<NP> ord(sz);
c5      REP(v, sz)
c1      for (auto it : nodes[v]->next) deg[it.second->id]++;
1e      ord[0] = nodes[0];
6e      int idx = 1;
b8      for (NP n : ord)
65        for (auto it : nodes[n->id]->next)
85          if (!--deg[it.second->id]) ord[idx++] = it.second;
15      swap(nodes, ord);
1c      REP(i, sz) nodes[i]->id = i;
24    }
0a  };
```

geometry

primitive.hpp

md5: 8fcbf7

```
7f  using D = double;
9b  const D PI = acos(D(-1)), EPS = 1e-10;
e0  int sgn(D a) { return (a < -EPS) ? -1 : (a > EPS); }
ac  int sgn(D a, D b) { return sgn(a - b); }
a2  struct P {
56    D x, y;
d9    P(D x = D(), D y = D()) : x(x), y(y) {}
83    P operator+(P r) const { return {x + r.x, y + r.y}; }
f7    P operator-(P r) const { return {x - r.x, y - r.y}; }
8b    P operator*(P r) const {
2e      return {x * r.x - y * r.y, x * r.y + y * r.x};
35    }
90    P operator*(D r) const { return {x * r, y * r}; }
0e    P operator/(D r) const { return {x / r, y / r}; }
ff    P& operator+=(P r) { return *this = *this + r; }
c2    P& operator-=(P r) { return *this = *this - r; }
97    P& operator*=(P r) { return *this = *this * r; }
e2    P& operator*=(D r) { return *this = *this * r; }
69    P& operator/=(D r) { return *this = *this / r; }
7c    P operator-() const { return {-x, -y}; }
8c    bool operator<(P r) const {
94      return 2 * sgn(x, r.x) + sgn(y, r.y) < 0;
34    }
cb    bool operator==(P r) const {
cf      return sgn((*this - r).rabs()) == 0;
ac    }
4b    bool operator!=(P r) const { return !(*this == r); }
b9    D norm() const { return x * x + y * y; }
f0    D abs() const { return sqrt(norm()); }
8f    D rabs() const {
8f      return max(std ::abs(x), std ::abs(y));
18    }
      // robust abs
19    D arg() const { return atan2(y, x); }
76    pair<D, D> to_pair() const { return {x, y}; }
da    static P polar(D le, D th) {
4d      return {le * cos(th), le * sin(th)};
b4    }
30  };
46  ostream& operator<<(ostream& os, const P& p) {
d9    return os << "P(" << p.x << ", " << p.y << ")";
d2  }
6d  struct L {
67    P s, t;
9c    L(P s = P(), P t = P()) : s(s), t(t) {}
03    P vec() const { return t - s; }
54    D abs() const { return vec().abs(); }
2b    D arg() const { return vec().arg(); }
4b  };
c3  ostream& operator<<(ostream& os, const L& l) {
be    return os << "L(" << l.s << ", " << l.t << ")";
fa  }
7d  D crs(P a, P b) { return a.x * b.y - a.y * b.x; }
28  D dot(P a, P b) { return a.x * b.x + a.y * b.y; }
    // cross(a, b) is too small?
19  int sgncrs(P a, P b) {
de    D cr = crs(a, b);
65    if (abs(cr) <= (a.rabs() + b.rabs()) * EPS) return 0;
19    return (cr < 0) ? -1 : 1;
b3  }
    // -2, -1, 0, 1, 2 : front, clock, on, cclock, back
f1  int ccw(P b, P c) {
e8    int s = sgncrs(b, c);
d4    if (s) return s;
25    if (!sgn(c.rabs()) || !sgn((c - b).rabs())) return 0;
32    if (dot(b, c) < 0) return 2;
c7    if (dot(-b, c - b) < 0) return -2;
c0    return 0;
b5  }
12  int ccw(P a, P b, P c) { return ccw(b - a, c - a); }
8f  int ccw(L l, P p) { return ccw(l.s, l.t, p); }
```

intersect.hpp

md5: 899d98

```
cb  P project(L l, P p) {
b3    P v = l.vec();
b5    return l.s + v * dot(v, p - l.s) / v.norm();
a4  }
79  bool insSL(L s, L l) {
76    int a = ccw(l, s.s), b = ccw(l, s.t);
6a    return a % 2 == 0 || b % 2 == 0 || a != b;
e4  }
34  bool insSS(L s, L t) {
3b    int a = ccw(s, t.s), b = ccw(s, t.t);
4b    int c = ccw(t, s.s), d = ccw(t, s.t);
60    return a * b <= 0 && c * d <= 0;
8d  }
cd  D distLP(L l, P p) {
ee    return abs(crs(l.vec(), p - l.s)) / l.abs();
69  }
42  D distSP(L s, P p) {
ae    P q = project(s, p);
40    if (ccw(s, q) == 0) return (p - q).abs();
18    return min((s.s - p).abs(), (s.t - p).abs());
e5  }
01  D distSS(L s, L t) {
03    if (insSS(s, t)) return 0;
40    return min({distSP(s, t.s), distSP(s, t.t),
47                distSP(t, s.s), distSP(t, s.t)});
89  }
```

cross_point.hpp

md5: a16c01

crossLL,crossSSの返り値は、1: 1点を共有、0: 共有点なし, -1: 無数の共有点(重なり)

```
d7  int crossLL(L l, L m, P& r) {
fa    D cr1 = crs(l.vec(), m.vec()),
27      cr2 = crs(l.vec(), l.t - m.s);
c9    if (sgncrs(l.vec(), m.vec()) == 0) {
16      r = l.s;
63      return sgncrs(l.vec(), l.t - m.s) ? 0 : -1;
0c    }
78    r = m.s + m.vec() * cr2 / cr1;
79    return 1;
ef  }
fa  int crossSS(L l, L m, P& r) {
f3    int u = crossLL(l, m, r);
1d    if (u == 0) return 0;
f9    if (u == -1) {
e9      r = max(min(l.s, l.t), min(m.s, m.t));
29      P q = min(max(l.s, l.t), max(m.s, m.t));
33      return (q < r) ? 0 : (q == r ? 1 : -1);
0e    }
a5    return ccw(l, r) == 0 && ccw(m, r) == 0 ? 1 : 0;
a1  }
```

polygon.hpp

md5: 78e152

insPolLは、多角形と直線の交点。sからtに向かう方向に並んでいる。直線からみると、多角形の内部が(すぐ左に見えるか)、(すぐ右に見えるか)の2つのプロパティがあり、それが切り替わる点一覧が返される。次の領域でのプロパティがsecondに2bitで表現される

```
6e  using Pol = V<P>;
a6  D area2(const Pol& pol) {
34    D u = 0;
48    if (!pol.size()) return u;
b9    P a = pol.back();
39    for (auto b : pol) u += crs(a, b), a = b;
5f    return u;
```

```
0e  }
    // 0: outside, 1: on line, 2: inside
51  int contains(const Pol& pol, P p) {
4e    if (!pol.size()) return 0;
8d    int in = -1;
c8    P _a, _b = pol.back();
ba    for (auto now : pol) {
b1      _a = _b;
d7      _b = now;
c9      P a = _a, b = _b;
be      if (ccw(a, b, p) == 0) return 1;
78      if (a.y > b.y) swap(a, b);
3f      if (!(a.y <= p.y && p.y < b.y)) continue;
5a      if (sgn(a.y, p.y) ? (crs(a - p, b - p) > 0)
9c                          : (a.x > p.x))
b7        in *= -1;
3b    }
cd    return in + 1;
b6  }

    // 以下 cross_point.hpp が必要

06  Pol convex_cut(const Pol& po, L l) {
23    if (!po.size()) return Pol{};
af    Pol q;
b3    P a, b = po.back();
79    for (auto now : po) {
af      a = b;
8c      b = now;
9f      if ((ccw(l, a) % 2) * (ccw(l, b) % 2) < 0) {
70        P buf;
c2        crossLL(l, L(a, b), buf);
f8        q.push_back(buf);
e5      }
26      if (ccw(l, b) != -1) q.push_back(b);
e7    }
4c    return q;
a4  }
    // (1:left) | (2: right) is inside between v[i] ~ v[i + 1]
ce  V<pair<P, int>> insPolL(Pol pol, L l) {
c7    using Pi = pair<P, int>;
de    V<Pi> v;
23    P a, b = pol.back();
ac    for (auto now : pol) {
a8      a = b;
67      b = now;
5e      P p;
41      if (crossLL({a, b}, l, p) != 1) continue;
ce      int sa = ccw(l, a) % 2, sb = ccw(l, b) % 2;
00      if (sa > sb) swap(sa, sb);
be      if (sa != 1 && sb == 1) v.push_back({p, 1});
94      if (sa == -1 && sb != -1) v.push_back({p, 2});
69    }
4a    sort(v.begin(), v.end(), [&](Pi x, Pi y) {
31      return dot(l.vec(), x.first - l.s) <
79             dot(l.vec(), y.first - l.s);
9f    });
8d    int m = int(v.size());
4b    V<Pi> res;
d6    for (int i = 0; i < m; i++) {
08      if (i) v[i].second ^= v[i - 1].second;
b1      if (!res.empty() && res.back().first == v[i].first)
bf        res.pop_back();
29      res.push_back(v[i]);
04    }
4b    return res;
78  }
```

circle.hpp

md5: 504af5

```
39  struct C {
b4    P p;
86    D r;
6a    C(P p = P(), D r = D()) : p(p), r(r) {}
aa  };
    // need Intersect/distLP, r.sはよりl.sに近い
a3  int crossCL(C c, L l, L& r) {
54    D u = distLP(l, c.p);
6d    int si = sgn(u, c.r);
17    if (si == 1) return 0;
87    P v = project(l, c.p);
27    P di =
6b        (si == 0)
04            ? P(0, 0)
77            : l.vec() * (sqrt(c.r * c.r - u * u) / l.abs());
ad    r = L(v - di, v + di);
7d    if (si == 0) return 1;
9c    return 2;
de  }
    // need Intersect/distLP, r.sはよりl.sに近い
d6  int crossCS(C c, L s, L& l) {
0b    if (!crossCL(c, s, l)) return 0;
7b    bool f1 = !ccw(s, l.s), f2 = !ccw(s, l.t);
2d    if (f1 && f2) return 2;
83    if (!f1 && !f2) return 0;
94    if (f1)
41      l.t = l.s;
48    else
11      l.s = l.t;
fb    return 1;
5c  }
    // return number of cross point
    // l, rはcから見た交点の角度、[l, r]がdに覆われている
c2  int crossCC(C c, C d, D& l, D& r) {
97    if (c.p == d.p) {
90      assert(sgn(c.r - d.r));
        // prohibit same circles
15      return 0;
de    }
1c    D di = (c.p - d.p).abs(), bth = (d.p - c.p).arg();
87    D costh =
52        (c.r * c.r + di * di - d.r * d.r) / (2 * c.r * di);
d6    int ty =
5c        min(sgn(c.r + d.r, di), sgn(di, abs(c.r - d.r)));
8d    if (ty == -1) return 0;
ad    if (ty == 0) costh = sgn(costh);
c7    D th = acos(max(D(-1), min(D(1), costh)));
88    l = bth - th;
cb    r = bth + th;
0d    return ty + 1;
8a  }
    // C(P(0, 0), r)とTri((0, 0), a, b)の共有面積
9e  D area2CT(C c, P _a, P _b) {
fc    P a = _a - c.p, b = _b - c.p;
03    D r = c.r;
99    if (a == b) return 0;
99    auto single = [&](P x, P y, bool tri) {
94      if (tri) return crs(x, y);
95      return r * r * ((y * P(x.x, -x.y)).arg());
09    };
02    bool ia = sgn(a.abs(), r) != 1,
92         ib = sgn(b.abs(), r) != 1;
8c    if (ia && ib) return single(a, b, true);
20    L l;
43    if (!crossCS(C(P(0, 0), r), L(a, b), l))
50      return single(a, b, false);
59    return single(a, l.s, ia) + single(l.s, l.t, true) +
94           single(l.t, b, ib);
bf  }
    // p, cの共有面積
05  D area2CPol(C c, const V<P>& po) {
f8    D sm = 0;
b4    P a, b = po.back();
17    for (auto p : po) {
88      a = b;
c1      b = p;
0e      sm += area2CT(c, a, b);
c9    }
5b    return sm;
50  }
```

circle_tangent.hpp

md5: abf1ee

```
    // pからcに接線を引く, 交点はp1, p2
2a  int tangent(C c, P p, P& p1, P& p2) {
21    D di = (c.p - p).abs();
dc    int si = sgn(c.r, di);
1d    if (si == 1) return 0;
cb    D th = acos(min(D(1), c.r / di));
e5    D ar = (p - c.p).arg();
36    p1 = c.p + P::polar(c.r, ar - th);
f5    p2 = c.p + P::polar(c.r, ar + th);
b8    if (si == 0) return 1;
06    return 2;
b7  }
    // 共通接線
```

```
c7  int tangent(C c, C d, L& l, L& r, bool inter) {
30    D di = (d.p - c.p).abs(), ar = (d.p - c.p).arg();
c8    if (sgn(di) == 0) {
dd      assert(sgn(c.r - d.r));
        // prohibit same circles
09      return 0;
76    }
aa    D costh = c.r + (inter ? d.r : -d.r);
3a    int si = sgn(abs(costh), di);
a4    costh /= di;
74    if (si == 1) return 0;
6e    if (si == 0) costh = sgn(costh);
f4    D th = acos(max(D(-1), min(D(1), costh)));
52    P base;
7a    base = P::polar(1, ar - th);
e7    l = L(c.p + base * c.r,
bf          d.p + base * d.r * (inter ? -1 : 1));
1a    base = P::polar(1, ar + th);
21    r = L(c.p + base * c.r,
64          d.p + base * d.r * (inter ? -1 : 1));
63    if (si == 0) return 1;
6f    return 2;
ab  }
```

circum_circle.hpp

md5: 3d9d89

```
7a  C circum_circle(P a, P b, P c) {
2b    b -= a;
a8    c -= a;
da    D s = 2 * crs(b, c);
3b    P x = {b.x, c.x};
08    P y = {b.y, c.y};
3f    P z = {b.norm(), c.norm()};
67    P r = P{crs(z, y), crs(x, z)} / s;
ec    return C(r + a, r.abs());
85  }
0a  C smallest_circle(V<P>& p, int i, array<P, 3> q, int j) {
78    if (i == int(p.size())) {
c7      if (j == 0) return C(P(0, 0), -1);
65      if (j == 1) return C(q[0], 0);
d8      if (j == 2)
3f        return C((q[0] + q[1]) / D(2.0),
f5                 (q[0] - q[1]).abs() / D(2.0));
52      if (j == 3) return circum_circle(q[0], q[1], q[2]);
19      assert(false);
d0    }
57    C c = smallest_circle(p, i + 1, q, j);
74    if (sgn((p[i] - c.p).abs(), c.r) == 1) {
7a      q[j] = p[i];
79      return smallest_circle(p, i + 1, q, j + 1);
4c    }
06    return c;
b9  }
3b  C smallest_circle(V<P> p) {
14    random_shuffle(begin(p), end(p));
f1    return smallest_circle(p, 0, array<P, 3>(), 0);
3d  }
```

convex_hull.hpp

md5: 6bb478

```
    // p must be sorted and uniqued!
16  V<P> convex_down(V<P> ps, bool on_edge = 0) {
75    assert(ps.size() >= 2);
dc    int r = -1, f = !on_edge;
f3    for (P d : ps) {
40      while (r > 0 && ccw(ps[r - 1], ps[r], d) % 2 < f) r--;
ea      ps[++r] = d;
ae    }
f4    ps.resize(r + 1);
71    return ps;
76  }
    // counter-clockwise
d9  V<P> convex(V<P> ps, bool on_edge = 0) {
23    sort(ps.begin(), ps.end());
42    ps.erase(unique(ps.begin(), ps.end()), ps.end());
6f    if (ps.size() <= 1) return ps;
e1    auto dw = convex_down(ps, on_edge);
ed    reverse(ps.begin(), ps.end());
3c    auto up = convex_down(ps, on_edge);
a7    dw.insert(dw.begin(), up.begin() + 1, up.end() - 1);
ee    return dw;
6b  }
```

convex_polygon_diameter.hpp

md5: 3fa6ae

```
2e  D diameter(const V<P>& p) {
ef    int n = int(p.size());
b8    if (n == 2) return (p[1] - p[0]).abs();
2a    int x = 0, y = 0;
3c    for (int i = 1; i < n; i++) {
4a      if (p[i] < p[x]) x = i;
4e      if (p[y] < p[i]) y = i;
a2    }
3b    D ans = 0;
63    int sx = x, sy = y;
      /*
      do {
      ...
      } while (sx != x || sy != y);
      で1周する
      */
ad    while (sx != y || sy != x) {
76      ans = max(ans, (p[x] - p[y]).abs());
ba      int nx = (x + 1 < n) ? x + 1 : 0,
0f          ny = (y + 1 < n) ? y + 1 : 0;
23      if (crs(p[nx] - p[x], p[ny] - p[y]) < 0)
93        x = nx;
da      else
99        y = ny;
85    }
eb    return ans;
3f  }
```

closest_pair.hpp

md5: 0f5400

```
9e  tuple<D, int, int> cp_rec(vector<pair<P, int>>& psi, int l,
41                            int r) {
39    if (l + 1 == r) return {1e100, -1, -1};
28    int m = midpoint(l, r);
12    D x = psi[m].first.x;
8c    auto dd = min(cp_rec(psi, l, m), cp_rec(psi, m, r));
33    auto [d, di, dj] = dd;
ba    inplace_merge(psi.begin() + l, psi.begin() + m,
95                  psi.begin() + r, [&](auto a, auto b) {
cf                    return a.first.y < b.first.y;
59                  });
1e    vector<int> pp;
6c    for (int a = l; a < r; a++) {
80      auto [p, i] = psi[a];
cd      if (p.x <= x - d || x + d <= p.x) continue;
f4      for (int b : pp | views::reverse) {
7d        auto [q, j] = psi[b];
2e        D dx = p.x - q.x, dy = p.y - q.y;
33        if (dy >= d) break;
08        D nd = sqrt(dx * dx + dy * dy);
4c        if (d > nd) d = nd, di = i, dj = j;
e2      }
73      pp.push_back(a);
89    }
30    return {d, di, dj};
ee  }
38  tuple<D, int, int> closest_pair(const vector<P>& ps) {
b2    int n = ssize(ps);
f3    vector<pair<P, int>> psi(n);
7a    REP(i, n) psi[i] = {ps[i], i};
93    sort(psi.begin(), psi.end(), [](auto a, auto b) {
62      return a.first.x < b.first.x;
fb    });
24    return cp_rec(psi, 0, n);
0f  }
```

halfplane_intersection.hpp

md5: c4d3af

```
34  int argcmp(P l, P r) {
9d    auto psgn = [&](P p) {
48      if (int u = sgn(p.y)) return u;
f5      if (sgn(p.x) == -1) return 2;
a9      return 0;
de    };
86    int lsgn = psgn(l), rsgn = psgn(r);
de    if (lsgn < rsgn) return -1;
00    if (lsgn > rsgn) return 1;
96    return sgncrs(r, l);
4d  }
    // line = (s,t) としたとき、 s-->t
```

```
   // の方向に見たときの右側を切り捨てる
78 V<L> halfplane_intersection(V<L> lines) {
ea   auto st = move(lines);
1b   sort(st.begin(), st.end(), [](L a, L b) {
bd     if (int u = argcmp(a.vec(), b.vec())) return u == -1;
d4     return sgncrs(a.vec(), b.s - a.s) < 0;
37   });
1c   st.erase(unique(st.begin(), st.end(),
ee                   [](L a, L b) {
41                     return !argcmp(a.vec(), b.vec());
57                   }),
e6            st.end());
38   ll l = 0, r = -1, err = 0;
3c   for (auto& g : st) {
6f     auto need = [&](L a, L b, L c) {
ad       D ab_dw = crs(a.vec(), b.vec()),
67         ab_up = crs(a.vec(), a.t - b.s);
eb       D bc_dw = crs(b.vec(), c.vec()),
81         bc_up = crs(c.t - b.s, c.vec());
18       if (ab_dw <= 0 || bc_dw <= 0) return true;
54       bool f = bc_up * ab_dw > bc_dw * ab_up;
67       if (!f && crs(a.vec(), c.vec()) < 0) err = 1;
5e       return f;
39     };
88     while (r - l >= 1 && !need(g, st[l], st[l + 1])) l++;
c7     while (r - l >= 1 && !need(st[r - 1], st[r], g)) r--;
68     if (r - l < 1 || need(st[r], g, st[l])) st[++r] = g;
99     if (err) return {};
35   }
72   if (r - l == 1 && !sgncrs(st[l].vec(), st[r].vec()) &&
9f     sgncrs(st[l].vec(), st[r].s - st[l].s) <= 0)
f5     return {};
22   return V<L>(st.begin() + l, st.begin() + r + 1);
c4 }
```

segment_ordering.hpp

md5: 05fd10

```
68 struct LI {
0d   L l;
40   ll i;
62   L reg() const { return l.s < l.t ? l : L{l.t, l.s}; }
5d   static ll d(P a, P b, P c) {
2a     return c < a || b < c ? 0 : sgncrs(a, b, c);
a3   }
8d   bool operator<(LI b) const {
c0     L l = reg(), r = b.reg();
3e     if (l.t < r.s || r.t < l.s) return 0;
18     ll f = 0;
52     f += d(l.s, l.t, r.s) - d(r.s, r.t, l.s);
88     f += d(l.s, l.t, r.t) - d(r.s, r.t, l.t);
30     if (f) return 0 < f;
af     return !(r.s < l.t) && (l.s < r.t || i < b.i);
d1   }
84 };
e0 V<ll> orderSegments(V<L> e, ll inf = 1ll << 30) {
0e   ll n = e.size(), f = 0;
33   for (auto& l : e)
23     if (l.t < l.s) swap(l.s, l.t);
1d   V<pair<P, int>> X(n * 2);
28   REP(i, n) X[i] = {e[i].s, i};
45   REP(i, n) X[n + i] = {e[i].t, n + i};
69   V<ll> d(n), res(n), ord(n, -1);
dc   V<V<ll>> adj(n);
67   std::set<LI> ds;
42   V<std::set<LI>::iterator> its(n, ds.end());
b6   ds.insert({{{-inf, -inf}, {inf, -inf}}, -1});
25   ds.insert({{{-inf, inf}, {inf, inf}}, -1});
f2   stable_sort(X.begin(), X.end());
5a   auto addEdge = [&](ll l, ll r) {
43     if (l >= 0 && r >= 0) adj[l].push_back(r), d[r]++;
fe   };
18   for (auto [x, i] : X) {
34     if (i < n) {
17       its[i] = ds.insert({e[i], i}).first;
11       auto l = its[i], r = its[i];
91       addEdge((--l)->i, i), addEdge(i, (++r)->i);
74     } else {
3a       i -= n;
7e       auto l = its[i], r = its[i];
45       addEdge((--l)->i, (++r)->i);
74       ds.erase(its[i]);
1f     }
7c   }
c5   REP(i, n) if (!d[i]) ord[f++] = i;
c8   for (ll v : ord)
7d     if (v >= 0)
f8       for (ll w : adj[v])
10         if (!--d[w]) ord[f++] = w;
e7   REP(i, n) res[ord[i]] = i;
47   return res;
05 }
```

other

hook_length_formula.hpp

md5: cf7510

```
1e Mint Hook_Length_Formula(vector<int>& A) {
97   assert(is_sorted(A.begin(), A.end()));
36   if (A.empty()) return Mint(1);
cc   vector<int> L(A.back(), 0);
ec   Mint n(1), d(1);
61   int s = 0;
bf   for (int l : A) REP(j, l) {
15       n *= Mint(++s);
f7       d *= Mint(L[j]++ + l - j);
f1     }
25   return n / d;
cf }
```

matrix.hpp

md5: a555f8

```
12 const Mint zero(0);
c7 const Mint one(1);
f9 bool iszero(Mint x) { return x.v == 0; }
20 template <class T>
03 struct Matrix {
0a   int H, W;
f6   V<V<T>> a;
81   Matrix(V<V<T>> v)
d8       : H(v.size()), W(v[0].size()), a(move(v)) {}
1c   Matrix(int H = 0, int W = 0)
a6       : H(H), W(W), a(H, V<T>(W)) {}
b7   static Matrix E(int n) {
c1     Matrix a(n, n);
20     REP(i, n) a.set(i, i, 1);
e5     return a;
1b   }
05   T at(int i, int j) const { return a[i][j]; }
7d   void set(int i, int j, T v) { a[i][j] = v; }
     /*
     左から var 列が階段行列になるように、行基本変形を施す。
     (e.g. 逆行列 は右に単位行列おいてから掃き出す)
     rank,determinant を返す。 in placeに更新されることに注意
     */
36   pair<int, T> sweep(int var) {
44     int i = 0, rank = 0;
60     T d = one;
3d     REP(j, var) {
12       for (i = rank; i < H && iszero(a[i][j]);) i++;
4a       if (i == H) continue;
31       if (i != rank) d = -d;
0e       swap(a[i], a[rank]);
3e       i = rank++;
24       T t = a[i][j];
eb       REP(k, W) a[i][k] /= t;
c6       d *= t;
d5       REP(k, H) if (k != i) {
fa         t = a[k][j];
90         REP(l, W) a[k][l] -= a[i][l] * t;
32       }
e6     }
e3     if (H != W || rank != H) d *= zero;
67     return {rank, d};
a7   }
2b   Matrix tr() {
df     auto B = Matrix(W, H);
da     REP(i, H) REP(j, W) B.set(j, i, at(i, j));
37     return B;
49   }
76   int getrank() {
d3     auto B = H < W ? *this : tr();
cb     return B.sweep(B.W).first;
b6   }
52   T getdet() { return Matrix(*this).sweep(W).second; }
12   Matrix inv() {
```

```
3a      assert(H == W);
6f      int N = H;
a1      Matrix X(N, 2 * N), B(N, N);
89      REP(i, N) REP(j, N) X.set(i, j, a[i][j]);
99      REP(i, N) X.set(i, i + N, one);
45      if (X.sweep(N).first < N) return Matrix();
e4      REP(i, N) REP(j, N) B.set(i, j, X.at(i, j + N));
6d      return B;
ab    }
80    friend ostream& operator<<(ostream& o, const Matrix& A) {
79      REP(i, A.H) {
99        REP(j, A.W) o << A.a[i][j] << " ";
3a        o << endl;
6a      }
3b      return o;
61    }
a5  };
```

surreal.hpp

md5: 266388

```
27  struct Surreal {
4c    using S = Surreal;
ac    using Int = long long;
13    Int p, q;  // val = p / 2**q
e2    Surreal(Int p = 0, Int q = 0) : p(p), q(q) {
f1      while (p % 2 == 0 && q) p /= 2, q--;
f8    }
39    S operator+(S r) const {
fe      auto c = max(q, r.q);
f4      return S((p << (c - q)) + (r.p << (c - r.q)), c);
eb    }
5d    S operator-() const { return {-p, q}; }
0c    bool operator<(S r) const { return (r + -*this).p > 0; }
2f    S lc() const {
99      if (!q && p <= 0) return p - 1;
31      return {p * 2 - 1, q + 1};
fb    }
65    friend S operator|(S l, S r) {
88      assert(l < r);
ab      S v = 0;
bf      while (!(l < v && v < r))
f1        v = l < v ? v.lc() : -(-v).lc();
2e      return v;
e2    }
26  };
```

memo

Nimber.md

$[(ab)_h, (ab)_l] = [(a_h \oplus a_l)(b_h \oplus b_l) \oplus a_l b_l, a_h b_h 2^{2^d - 1} \oplus a_l b_l]$

$[(\mathrm{inv}\ x)_h, (\mathrm{inv}\ x)_l] = \mathrm{inv}((x_h \oplus x_l)x_l \oplus x_h x_h 2^{2^d-1})[x_h, x_h \oplus x_l]$

ラグランジュの反転公式.md

F, G が多項式で $F(G(x)) = G(F(x)) = x$ のとき

$[x^n]G(x)^k = \frac{k}{n}[x^{-k}]F(x)^{-n}$

$[x^n]H(G(x)) = \frac{1}{n}[x^{-1}]H'(x)F(x)^{-n}$

二部グラフ.md

- 最小点被覆 : L のマッチされてない頂点から、到達不能な L の頂点と到達可能な R の頂点。
- 最大独立点集合 : 最小点被覆の補集合
- 最小辺被覆 : 最大マッチングに加えて、マッチングで覆われていない各頂点に接続する辺を 1 つずつ選ぶ。

五心.md

$f(w_a, w_b, w_c) = (w_a\vec{a} + w_b\vec{b} + w_c\vec{c})/(w_a + w_b + w_c)$ として、

- 外心 $\vec{o} = f(\sin 2A, \sin 2B, \sin 2C)$
- 内心 $\vec{i} = f(|\vec{a}|, |\vec{b}|, |\vec{c}|)$（引数の符号を 1 個変えると傍心）
- 垂心 $\vec{h} = f(\tan A, \tan B, \tan C)$
- 重心 $\vec{g} = f(1, 1, 1) = (2\vec{o} + \vec{h})/3$

数え上げ公式.md

オイラーグラフについて、 Eulerian Cycle の個数は $C_v \times \prod_{w \in V}(\mathrm{outdeg}_w - 1)!$ (C_v は v を根とする有向全域木の個数、 v に依存しない)

v に向かう有向全域木の個数は、次の行列の v 行目と v 列目を除いた行列の行列式に等しい： 辺 $u \to v$ に対して a_{uv}-=1 , a_{uu}+=1 .

整数.md

二項係数

$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$

$\binom{n}{k} = \frac{n}{k}\binom{n-1}{k-1}$

$\binom{L}{k} + \cdots + \binom{R-1}{k} = \binom{R}{k+1} - \binom{L}{k+1}$

第一種スターリング数

$c(n, k)$: $1, 2, \ldots, n$ の順列で巡回置換 k 個に分割できるものの個数

$c(n, k) = c(n-1, k-1) + (n-1)c(n-1, k)$

母関数 (k) : $\sum_{k=0}^{N} x^k c(N, k) = \prod_{i=0}^{N-1}(x - i)$

指数型母関数 (n) : $\sum_{n=0}^{\infty}(x^n/n!)c(n, K) = (\log(1+x))^k/K!$

第二種スターリング数

$S(n, k)$: $1, 2, \ldots, n$ を k 個の区別しない集合に分割する方法の数

$S(n, k) = S(n-1, k-1) + kS(n-1, k)$

$x^n = \sum_{k=0}^{n} S(n, k)x(x-1)\ldots(x-k+1)$

指数型母関数 (n) : $\sum_{n=0}^{\infty}(x^n/n!)S(n, K) = (e^x - 1)^K/K!$

母関数 (k) : $\sum_{k=0}^{N} x^k S(N, k) = \left(\sum_{i=0}^{\infty}(-1)^i x^i/i!\right)\left(\sum_{i=0}^{\infty} i^N x^i/i!\right)$

ベル数

B_n: $1, 2, \ldots, n$ をいくつかの集合に分割する方法の数

n	0	1	2	3	4	5	6	7	8	9	10
B_n	1	1	2	5	15	52	203	877	4140	21147	115975

$B_{n+1} = \sum_{k=0}^{n}\binom{n}{k}B_k$

$B_n = e^{-1}\sum_{k=0}^{\infty} k^n/k!$

指数型母関数 : $\exp(\exp x - 1) = \sum_{n=0}^{\infty} B_n(x^n/n!)$

カタラン数

C_n: n 個の (と) を括弧列になるように並べる方法の数

n	0	1	2	3	4	5	6	7	8	9	10
C_n	1	1	2	5	14	42	132	429	1430	4862	16796

$C_n = \frac{1}{n+1}\binom{2n}{n} = \frac{(2n)!}{(n+1)!n!}$

$C_{n+1} = \frac{2(2n+1)}{n+2}C_n = \sum_{k=0}^{n} C_k C_{n-k}$

母関数の累乗 : $C(x)^k = ((1 - \sqrt{1-4x})/2x)^k = \sum_{n=0}^{\infty}\frac{k}{n+k}\binom{2n+k-1}{n}x^n$

モンモール数

a_n: $1, 2, \ldots, n$ の順列 P で $P_i \neq i$ となるものの個数

n	0	1	2	3	4	5	6	7	8	9	10
a_n		0	1	2	9	44	265	1854	14833	133496	1334961

$a_n = na_{n-1} + (-1)^n$

指数型母関数 : $\sum_{n=0}^{\infty} a_n x^n / n! = e^{-x}/(1-x)$

分割数

P_n: n を正の整数の和として表す方法の数

n	0	1	2	3	4	5	6	7	8	9	10		20	30	40	50
P_n	1	1	2	3	5	7	11	15	22	30	42		627	5604	37338	204226

母関数 : $\left(\sum_{n=-\infty}^{\infty} (-1)^n x^{n(3n+1)/2}\right)^{-1}$

ベルヌーイ数

指数型母関数 : $\sum_{0 \le n} (x^n/n!) B_n = x/(e^x - 1)$

$1^K + 2^K + \cdots + N^K = K! \cdot \sum_{j=0}^{K} (B_j/j!)(N^{K-j+1}/(K-j+1)!)$

母関数.md

$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots = \sum_{n=0}^{\infty} \frac{x^n}{n!}$

$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} x^n$

$\sqrt{1-x} = 1 - \frac{1}{2}x - \frac{1}{8}x^2 - \frac{1}{16}x^3 - \frac{5}{128}x^4 - \cdots = 1 - \sum_{n=1}^{\infty} \frac{(2n-2)!}{2^{2n-1} n!(n-1)!} x^n$

$\frac{1}{\sqrt{1-x}} = 1 + \frac{1}{2}x + \frac{3}{8}x^2 + \frac{5}{16}x^3 + \frac{35}{128}x^4 - \cdots = \sum_{n=0}^{\infty} \frac{(2n)!}{4^n n!} x^n$

燃やす埋める.md

条件	処理
i が 0 のとき c 失う	`make_edge(i, G, c)`
i が 1 のとき c 失う	`make_edge(S, i, c)`
i が 0 のとき c 得る	先に c を得る `make_edge(S, i, c)`
i が 1 のとき c 得る	先に c を得る `make_edge(i, G, c)`
i が $0, j$ が 1 のとき c 失う	`make_edge(i, j, c)`
i が $1, j$ が 0 のとき c 失う	`make_edge(j, i, c)`
i と j が異なるとき c 失う	`make_edge(i, j, c)` `make_edge(j, i, c)`

i が $0, j$ が 0 のとき c 得る	先に c を得る
	`make_edge(S, k, c)`
	`make_edge(k, i, inf)`
	`make_edge(k, j, inf)`

i が $1, j$ が 1 のとき c 得る	先に c を得る
	`make_edge(i, k, inf)`
	`make_edge(j, k, inf)`
	`make_edge(k, G, c)`

素数.md

素数の個数

n	10^2	10^3	10^4	10^5	10^6	10^7	10^8	10^9	10^{10}
$\pi(n)$	25	168	1229	9592	78498	664579	5.76e+6	5.08e+7	4.55e+8

高度合成数

$\le n$	10^3	10^4	10^5	10^6	10^7	10^8	10^9
x	840	7560	83160	720720	8648640	73513440	735134400
$d^0(x)$	32	64	128	240	448	768	1344

$\le n$	10^{10}	10^{11}	10^{12}	10^{13}	10^{14}	10^{15}	10^{16}	10^{17}	10^{18}
$d^0(x)$	2304	4032	6720	10752	17280	26880	41472	64512	103680

素数階乗

n	2	3	5	7	11	13	17	19	23	29
$n\#$	2	6	30	210	2310	30030	510510	9.70e+6	2.23e+8	6.47e+9

階乗

$4!$	$5!$	$6!$	$7!$	$8!$	$9!$	$10!$	$11!$	$12!$	$13!$
24	120	720	5040	40320	362880	3.63e+6	3.99e+7	4.79e+8	6.23e+9